

## Global Facilities - Design & Construction Standard - Gas - Cylinder Spec Gas & Bulk Spec Gas Supply System (CSG, BSGS)

|                |                   |                    |                           |
|----------------|-------------------|--------------------|---------------------------|
| ECN Facility:  | GLOBAL FACILITIES | ECN Area:          | ENGINEERING               |
| Owner(s)       | ENGINEERING       | Target Audience:   | All Facilities            |
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[ECN History](#)[Submit a Change Request](#)

### 1.0 Purpose

- 1.1 This document serves as the System Standard that defines the minimum requirements for the design, engineering, fabrication, installation, test and commissioning for new Cylinder Specialty Gas and Bulk Specialty Gas Supply Systems as part of new Greenfield FAB Construction projects.
- 1.2 The standard is applicable for supply of specialty gases from cylinders or larger containers (e.g., cylinder bundles, Y-cylinders, tonners, tube trailers, ISO containers, etc.) within Gas Cabinets/Racks and Bulk Specialty Gas Systems (BSGS).
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects and purchase specifications for purchasing new spec gas equipment.
- 1.4 For sites that require higher standards, specifications, or regulations for compliance, etc., those sites may have to follow their existing (or better) standards, specifications, or regulations in order to meet site specific requirements. All deviations from this System Standard/Specification shall be documented by the site team including justification and shall be reviewed and approved by Corporate GFTT and FCT. Refer to Appendix 1 for Template Form for Site Specific Deviation List.
  - a. During the design development stage of a new, greenfield construction project, the vendor, contractor, and/or project team

shall review the Global System Standard and create the list of proposed deviations.

- b. Project team shall review and perform a risk assessment for each deviation using RPN SOD criteria, then obtain approval from Project Manager.
- c. The Deviation list shall be reviewed and approved by all stakeholders (GFTT, FCT, MCT, EHS, FAB Support, AMHS, etc.) prior to Global Facilities (GFTT & FCT) Directors' review.
- d. After approval from Global Facilities Directors, the deviation list shall be reviewed and approved by Global Construction and Site Facilities Directors before authorization to proceed.

## 2.0 Scope

2.1 The document covers new Basebuild System installations, and does NOT cover tool installation. This document provides for:

- Design and specification for purchase of cylinder specialty gas and bulk specialty gas supply equipment.
- Design of specialty gas rooms and utility requirements

2.2 For existing site improvements/upgrades or additions to existing gas system installations at existing sites, consult FCT for Standard System/Specification to follow.

2.3 Site(s) impacted:

- a. This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield Fab Construction projects.
- b. Based upon requirements and applicability with existing design, this document may also be used for Brownfield or other expansion projects upon review with FCT.

### 3.0 Definitions and Acronyms

| Item                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Boiling Point</b>                           | The temperature at which the vapor pressure of a liquid equals the atmospheric pressure of 14.7 psia (101 kPa) or 760 mmHg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Bulk Specialty Gas Supply System (BSGS)</b> | A supply system connected to a bulk source (e.g., cylinder bundle, Y-cylinder, tonner, or tube trailer) of a specialty gas, consisting of its associated gas panel, and connected to gas distribution piping                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Class I, Division 2 Location</b>            | Location in which:<br>(1) volatile flammable gases, flammable liquid-produced vapors, or combustible liquid-produced vapors are handled, processed, or used, but in which the liquids, vapors, or gases will normally be confined within closed containers or closed systems from which they can escape only in case of accidental rupture or breakdown of such containers or systems or in case of abnormal operation of equipment, or<br>(2) ignitable concentrations of flammable gases, flammable liquid-produced vapors, or combustible liquid-produced vapors are normally prevented by positive mechanical ventilation and which might become hazardous through failure or abnormal operation of the ventilating equipment, or<br>(3) adjacent to a Class I, Division 1 location, and to which flammable liquid-produced vapors, or combustible liquid-produced vapors above their flash points might occasionally be communicated unless such communication is prevented by adequate positive-pressure ventilation from a source of clean air and effective safeguards against ventilation failure are provided. |
| <b>Closed System</b>                           | The use of a solid or liquid hazardous material and all uses of compressed gases involving a closed vessel or system that remains closed during normal operations where vapors emitted by the product are not liberated outside of the vessel or system during normal operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Compressed Gas</b>                          | A material, or mixture of materials that:<br>1) Is a gas at 68°F (20°C) or less at 14.7 psia (101 kPa) of pressure; and<br>2) Has a boiling point of 68°F (20°C) or less at 14.7 psia (101 kPa) which is either liquefied, non-liquefied or in solution, except those gases which have no other health- or physical-hazard properties are not considered to be compressed until the pressure in the packaging exceeds 41 psia (282 kPa) at 68°F (20°C).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| <b>Item</b>                     | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Compressed Gas Container</b> | A pressure vessel designed to hold compressed gases at pressures greater than one atmosphere at 68°F (20°C) and includes cylinders, containers, and tanks.                                                                                                                                                                                                                                                                                                          |
| <b>Control Area</b>             | Spaces within a building where quantities of hazardous materials not exceeding the maximum allowable quantities per control area are stored, dispensed, used or handled.                                                                                                                                                                                                                                                                                            |
| <b>Corrosive</b>                | A chemical that causes visible destruction of, or irreversible alterations in, living tissue by chemical action at the point of contact.                                                                                                                                                                                                                                                                                                                            |
| <b>Cylinder</b>                 | A pressure vessel designed for pressures higher than 40 psia (275.6 kPa) and having a circular cross section. It does not include a portable tank, multiunit tank car tank, cargo tank or tank car.                                                                                                                                                                                                                                                                 |
| <b>EFS</b>                      | Excess Flow Switch                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Emergency Power System</b>   | A source of automatic electric power of a required capacity and duration to operate required life safety, fire alarm, detection and ventilation systems in the event of a failure of the primary power. Emergency power systems are required for electrical loads where interruption of the primary power could result in loss of human life or serious injuries.                                                                                                   |
| <b>Excess Flow Control</b>      | A fail-safe system or other approved means designed to shut off flow caused by a rupture in pressurized piping systems.                                                                                                                                                                                                                                                                                                                                             |
| <b>Exhausted Enclosure</b>      | An appliance or piece of equipment which consists of a top, a back, and two sides providing a means of local exhaust for capturing gases, fumes, vapors, and mists. Such enclosures include laboratory hoods, exhaust fume hoods and similar appliances and equipment used to retain and exhaust locally the gases, fumes, vapors, and mists that could be released. Rooms or areas provided with general ventilation, in themselves, are not exhausted enclosures. |
| <b>FMCS</b>                     | Facility Monitoring and Control System                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Fabrication Area</b>         | An area within a semiconductor fabrication facility and related research and development areas in which there are processes using hazardous production materials. Such areas are allowed to include ancillary rooms or areas such as dressing rooms and offices that are directly related to the fabrication area processes.                                                                                                                                        |
| <b>Fail-Safe</b>                | A design condition incorporating a feature for automatically counteracting the effect of an anticipated possible source of failure; also, a design condition eliminating or mitigating a hazardous condition by compensating automatically for a failure or malfunction.                                                                                                                                                                                            |

| <b>Item</b>                    | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Failure</b>                 | the termination of the ability of an item to perform a required function. Failure is an event, as distinguished from 'fault,' which is a state.                                                                                                                                                                                                                                                                                                                                                      |
| <b>Fault</b>                   | the state of an item characterized by inability to perform a required function, excluding the inability during preventive maintenance or other planned actions, or due to lack of external resources.                                                                                                                                                                                                                                                                                                |
| <b>Flammable Gas</b>           | A compressed gas which (1) is ignitable at 14.7 psia (101 kPa) when in mixture of 13 percent or less by volume with air; or (2) has a flammable range at 14.7 psia (101 kPa) with air of not less than 12 percent, regardless of the lower limit.                                                                                                                                                                                                                                                    |
| <b>Flammable Liquefied Gas</b> | A liquefied compressed gas which, under a charged pressure, is partially liquid at a temperature of 68°F (20°C) and which is flammable.                                                                                                                                                                                                                                                                                                                                                              |
| <b>Flammable Liquid</b>        | A liquid having a closed cup flash point below 100°F (38°C). Flammable liquids are further categorized into a group known as Class I liquids:<br>1) <b>Class IA</b> – Liquids having a flash point below 73°F (23°C) and having a boiling point below 100°F (38°C).<br>2) <b>Class IB</b> – Liquids having a flash point below 73°F (23°C) and having a boiling point at or above 100°F (38°C).<br>3) <b>Class IC</b> – Liquids having a flash point at or above 73°F (23°C) and below 100°F (38°C). |
| <b>Flame Arrestor</b>          | Device to stop a flame front in case of acetylene decomposition                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Flashback Arrestor</b>      | Device to stop a flame front and the flow of gas in the case of acetylene decomposition. This device can be activated either by a pressure shock wave or by a temperature sensing device.                                                                                                                                                                                                                                                                                                            |
| <b>Gas Cabinet</b>             | A full enclosed, ventilated, noncombustible enclosure used to provide an isolated environment for compressed gas cylinders in storage or use. Doors and access ports for exchanging cylinders and accessing pressure-regulating controls are allowed to be included.                                                                                                                                                                                                                                 |
| <b>Gas Panel</b>               | an arrangement of fluid handling components (e.g., valves, filters, mass flow controllers) that regulates the flow of fluids into the process. Synonyms: gas jungle, jungle, gas control valves, valve manifold.                                                                                                                                                                                                                                                                                     |
| <b>Gas Panel Enclosure</b>     | an enclosure designed to contain leaks from gas panel(s) within itself. Synonyms: jungle enclosure, gas box, valve manifold box.                                                                                                                                                                                                                                                                                                                                                                     |

| Item                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Gas Rack</b>                            | open rack structure used to support inert gas cylinders, and its associated gas panel, and connected to gas distribution piping or to equipment using the gas.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Gas Room</b>                            | A separately ventilated, fully enclosed room in which only compressed gases and associated equipment and supplies are stored or used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>HPM Room</b>                            | A room used in conjunction with or serving a Group H-5 occupancy, where HPM is stored or used and which is classified as a Group H-2, H-3 or H-4 occupancy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hazardous Production Material (HPM)</b> | a solid, liquid, or gas associated with semiconductor manufacturing that has a degree-of-hazard rating in health, flammability, or instability of class 3 or 4 as ranked by NFPA 704, and which is used directly in research, laboratory, or production processes which have as their end product, materials that are not hazardous.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Health Hazard</b>                       | A classification of a chemical for which there is statistically significant evidence that acute or chronic health effects are capable of occurring in exposed persons. The term "health hazard" includes chemicals that are toxic, highly toxic and corrosive.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Highly Toxic</b>                        | A material which produces a lethal dose or lethal concentration which falls within any of the following categories:<br>1) A chemical that has a median lethal dose (LD50) of 50 milligrams or less per kilogram of body weight when administered orally to albino rats weighing between 200 and 300 grams each.<br>2) A chemical that has a median lethal dose (LD50) of 200 mg or less per kg of body weight when administered by continuous contact for 24 hours (or less if death occurs within 24 hours) with the bare skin of albino rabbits weighing between 2 and 3 kg each.<br>3) A chemical that has a medium lethal concentration (LC50) in air of 200 parts per million by volume or less of gas or vapor, or 2 milligrams per liter or less of mist, fume or dust, when administered by continuous inhalation for one hour (or less if death occurs within 1 hour) to albino rats weighing between 200 and 300 grams each. |
| <b>Highly Volatile Liquid</b>              | A liquefied compressed gas with a boiling point of less than 68°F (20°C).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>High Hazard Group H-5 Occupancy</b>     | an occupancy classification which includes, among others, semiconductor fabrication facilities and comparable research and development areas in which hazardous production materials (HPM) are used and the aggregate quantity of materials is in excess of those listed in Tables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Item                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|                                                        | 5003.1.1(1) and 5003.1.1(2) as provided in IFC Chapter 50 "Hazardous Materials – General Provisions".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Immediately Dangerous to Life and Health (IDLH)</b> | The concentration of airborne contaminants that poses a threat of death, immediate or delayed permanent adverse health effects, or effects that could prevent escape from such an environment as established by the National Institute of Occupational Safety and Health (NIOSH) based on both toxicity and flammability.                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Incompatible Materials</b>                          | Materials that, when mixed, have the potential to react in a manner which generates heat, fumes, gases or byproducts which are hazardous to life or property.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Inert Gas</b>                                       | A gas that is capable of reacting with other materials only under abnormal conditions such as high temperatures, pressures and similar extrinsic physical forces. Inert gases do not exhibit either physical or health hazard properties as defined (other than acting as a simple asphyxiant) or hazard properties other than those of a compressed gas.                                                                                                                                                                                                                                                                                                                                           |
| <b>Interlock</b>                                       | a mechanical, electrical or other type of device or system, the purpose of which is to prevent or interrupt the operation of specified machine elements under specified conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>LOTO/COHE</b>                                       | Lockout Tagout / Control of Hazardous Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Local Exhaust Ventilation (LEV)</b>                 | a system that captures contaminant at or near its source and moves the contaminant to the external environment, usually through an air cleaning or a destructive device. It is not to be confused with laminar flow ventilation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Lower Flammable Limit (LFL)</b>                     | The minimum concentration of vapor in air at which propagation of flame will occur in the presence of an ignition source. The LFL is sometimes referred to a LEL or lower explosive limit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>N+1+F</b>                                           | System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, +1 future unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Oxidizer</b>                                        | <p>A material that readily yields oxygen or other oxidizing gas, or that readily reacts to promote or initiate combustion of combustible materials and, if heated or contaminated, can result in vigorous self-sustained decomposition.</p> <ol style="list-style-type: none"> <li>1) Class 1 – an oxidizer that does not moderately increase the burning rate of combustible materials.</li> <li>2) Class 2 – an oxidizer that will cause a moderate increase in the burning rate of combustible materials with which it comes in contact.</li> <li>3) Class 3 – an oxidizer that causes a severe increase in the burning rate of combustible materials with which it comes in contact.</li> </ol> |

| Item                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                                           | 4) Class 4 – an oxidizer that can undergo an explosive reaction due to contamination or exposure to thermal or physical shock and that causes a severe increase in the burning rate of combustible materials with which it comes into contact.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Oxidizing Gas</b>                      | A gas that can support and accelerate combustion of other materials more than air does.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>PLC</b>                                | Programmable Logic Controller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>POC/LPOC</b>                           | Point of Connection / Lateral Point of Connection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Pyrophoric</b>                         | a chemical with an autoignition temperature in air, at or below a temperature of 54.4°C (130°F).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>SCADA</b>                              | Supervisory Control and Data Acquisition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Semiconductor Fabrication Facility</b> | a building or a portion of a building in which electrical circuits or devices are created on solid crystalline substances having electrical conductivity greater than insulators but less than conductors. These circuits or devices are commonly known as semiconductors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Standby Power System</b>               | a source of automatic electric power of a required capacity and duration to operate required building, hazardous materials or ventilation systems in the event of a failure of the primary power. Standby power systems are required for electrical loads where interruption of the primary power could create hazards or hamper rescue or fire-fighting operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Supplemental Exhaust</b>               | local exhaust ventilation that is used intermittently for a specific task of finite duration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Toxic</b>                              | A chemical falling within any of the following categories:<br>1) A chemical that has a median lethal dose (LD50) of more than 50 milligrams per kilogram, but not more than 500 milligrams per kilogram of body weight when administered orally to albino rats weighing between 200 and 300 grams each.<br>2) A chemical that has a median lethal dose (LD50) of more than 200 milligrams per kilogram but not more than 1,000 milligrams per kilogram of body weight when administered by continuous contact for 24 hours (or less if death occurs within 24 hours) with the bare skin of albino rabbits weighing between 2 and 3 kilograms each.<br>3) A chemical that has a medium lethal concentration (LC50) in air of more than 200 parts per million but not more than 2,000 parts per million by volume of gas or vapor, or more than 2 milligrams per liter but not more than 20 milligrams per liter of mist, fume or dust, when administered by continuous inhalation for 1 hour (or less |



| Item                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | if death occurs within 1 hour) to albino rats weighing between 200 and 300 grams each.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>TUM</b>                          | Tool Utility Matrix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>UPS</b>                          | Uninterruptible Power Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Unstable (Reactive) Material</b> | <p>A material, other than an explosive, which in the pure state or as commercially produced, will vigorously polymerize, decompose, condense or become self-reactive and undergo other violent chemical changes, including explosion, when exposed to heat, friction, or shock, or in the absence of an inhibitor, or in the presence of contaminants, or in contact with incompatible materials.</p> <ol style="list-style-type: none"> <li>1) Class 1 – Materials that in themselves are normally stable but which can become unstable at elevated temperatures and pressures.</li> <li>2) Class 2 – Materials that in themselves are normally unstable and readily undergo violent chemical change but do not detonate.</li> <li>3) Class 3 – Materials that in themselves are capable of detonation or of explosive decomposition or explosive reaction but which require a strong initiating source or which must be heated under confinement before initiation.</li> <li>4) Class 4 – Materials that in themselves are readily capable of detonation or explosive decomposition or explosive reaction at normal temperatures and pressures.</li> </ol> |
| <b>Ventilation</b>                  | The natural or mechanical process of supplying conditioned or unconditioned air to, or removing such air from, any space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Water-Reactive Material</b>      | <p>A material that explodes; violently reacts; produces flammable, toxic or other hazardous gases; or evolves enough heat to cause autoignition or ignition of combustibles upon exposure to water or moisture.</p> <ol style="list-style-type: none"> <li>1) Class 1 – Materials that react with water with some release of energy, but not violently.</li> <li>2) Class 2 – Materials that react violently with water or have the ability to boil water. Materials that produce flammable, toxic or other hazardous gases, or evolve enough heat to cause autoignition or ignition of combustibles upon exposure to water or moisture.</li> <li>3) Class 3 – Materials that react explosively with water without requiring heat or confinement.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Z-Purge</b>                      | The process of supplying an enclosure with a protective gas with or without continuous flow at sufficient pressure to prevent the entrance of a flammable or corrosive gas or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Global Facilities - Design &amp; Construction Standard - Gas - Cylinder Spec Gas &amp; Bulk Spec Gas Supply System

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| Item                                      | Description                                                                                                                                                                                                              |
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|                                           | vapor. Type Z pressurization reduces the classification within an enclosure from Division 2 to nonhazardous.                                                                                                             |
| FCT                                       | Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs                                                                                                 |
| GFTT                                      | Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards                                                                               |
| GFC                                       | Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities |
| Global Facilities Central Team            | This is used interchangeably with Innovation & Integration Team                                                                                                                                                          |
| Global Facilities Construction Team       | This is used interchangeably with Global Construction Team                                                                                                                                                               |
| Global Facilities Operations Support Team | This is used interchangeably with Global Facilities Operations Team                                                                                                                                                      |
| Global Facilities Planning Team           | This is used interchangeably with Global Facilities Line Planning Team                                                                                                                                                   |

## 4.0 Roles and Responsibilities

| Role                                    | Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global Facilities (GFTT)                | <ul style="list-style-type: none"> <li>• Develop and maintain system standard document.</li> <li>• Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard.</li> <li>• Perform periodic document review and update System Standard.</li> <li>• Support site teams by providing knowledge and technical expertise.</li> <li>• Review site specific deviations from this system standard.</li> </ul>                                                                                                                                                                                |
| Global Facilities (FCT)                 | <ul style="list-style-type: none"> <li>• Review and provide technical feedback on System Standard.</li> <li>• Update BKMs and Develop relevant BKMs.</li> <li>• Support the site teams by providing knowledge and technical expertise in operation and maintenance.</li> <li>• Review site specific deviations from this system standard.</li> </ul>                                                                                                                                                                                                                                                                                    |
| Site Project Construction Teams         | <ul style="list-style-type: none"> <li>• Review and provide technical feedback on System Standard</li> <li>• Incorporate System Standard into Project Engineering, Design, and Construction</li> <li>• Document project deviations, exceptions, and exclusions from System Standard</li> <li>• Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard.</li> <li>• Consult with Global/Site PSM on the Relative Ranking Analysis (RRA) and ensure the RRA is less than 30. Otherwise, carry out mitigation till RRA comes below 30.</li> </ul> |
| Site Facilities Team                    | <ul style="list-style-type: none"> <li>• Use System Standard as technical reference for new Greenfield projects.</li> <li>• Identify and document deviations from the system standard/specification for review and approval by GFTT and FCT.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 |
| Equipment Supplier or other contractors | <ul style="list-style-type: none"> <li>• Meet or exceed the requirements as documented in Micron standards.</li> <li>• Identify and justify non-compliance or deviations from system standard for which supplier or contractor is unable to meet, and recommend alternate solution for review by Site Team, GFTT, and FCT.</li> </ul>                                                                                                                                                                                                                                                                                                   |
| Global/Site PSM                         | <ul style="list-style-type: none"> <li>• Provide guidance for Relative Ranking Analysis rating for systems.</li> <li>• Consult on required mitigation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 5.0 Codes, Standards, and References

Cylinder Specialty Gas and Bulk Specialty Gas Supply Systems shall be designed according to NFPA, SEMI, CGA, and other applicable local codes and regulations and/or adopted equivalent standards. The following guidelines and codes in the table below shall be applied appropriately.

If the local code requirements are stricter than any of the codes/standards listed in the below table, the stricter codes shall be followed and noted in the deviations list. Otherwise, compliance to the referenced codes and standards shall apply.

Contractors shall submit written notice to Micron of variances or deviations between the Contract documents and legal requirements.

The codes, standards and references in Table 1 were used in the writing of this design standard.

**Table 1 – Codes, Standards, and References**

Table 1 below specifies relevant codes and standards for different disciplines.

| Standard/ Code/ Reference |                            | Description                                                                        |
|---------------------------|----------------------------|------------------------------------------------------------------------------------|
| FM Global                 | 7-7 / 17-12<br>(July 2020) | Semiconductor Fabrication Facilities                                               |
| FM Global                 | 1-2<br>(April 2018)        | Earthquakes                                                                        |
| FM Global                 | 1-11<br>(July 2016)        | Fire Following Earthquake                                                          |
| FM Global                 | 1-23<br>(April 2012)       | Fire Barriers and Protection of Openings                                           |
| FM Global                 | 1-44<br>(April 2012)       | Damage-Limiting Construction                                                       |
| IBC                       | 2015                       | International Building Code (IBC)                                                  |
| IFC                       | 2018                       | International Fire Code (IFC)                                                      |
| NFPA                      | 55<br>(2016)               | Compressed Gases and Cryogenic Fluids Code                                         |
| NFPA                      | 70<br>(2017)               | National Electrical Code (NEC)                                                     |
| NFPA                      | 110<br>(2019)              | Standard for Emergency and Standby Power Systems                                   |
| NFPA                      | 318<br>(2018)              | Standard for Protection of Semiconductor Fabrication Facilities                    |
| NFPA                      | 704<br>(2017)              | Standard for the Identification of the Hazards of Materials for Emergency Response |

| Standard/ Code/ Reference |                | Description                                                                                                           |
|---------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------|
| OSHA                      | 29 CFR 1910    | United States Occupational Safety and Health Administration (OSHA)                                                    |
| SEMI                      | E49-0419       | Guide for High Purity and Ultrahigh Purity Piping Performance, Subassemblies, and Final Assemblies                    |
| SEMI                      | E49.6-1103     | Guide for Subsystem Assembly and Testing Procedures – Stainless Steel Systems                                         |
| SEMI                      | F1-0812        | Specification for Leak Integrity of High-Purity Gas Piping Systems and Components                                     |
| SEMI                      | F78-0611       | Practice for Gas Tungsten Arc (GTA) Welding of Fluid Distribution Systems in Semiconductor Manufacturing Applications |
| SEMI                      | S2-0818        | Environmental, Health, and Safety Guideline for Semiconductor Manufacturing Equipment                                 |
| SEMI                      | S5-0616        | Safety Guideline for Sizing and Identifying Flow Limiting Devices for Gases                                           |
| SEMI                      | S6-0618        | Environmental, Health, and Safety Guideline for Exhaust Ventilation of Semiconductor Manufacturing Equipment          |
| SEMI                      | S7-0515        | Safety Guidelines for Environmental, Safety, and Health (ESH) Evaluation of Semiconductor Manufacturing Equipment     |
| SEMI                      | S8-0218        | Safety Guidelines for Ergonomics Engineering of Semiconductor Manufacturing Equipment                                 |
| SEMI                      | S10-1119       | Safety Guideline for Risk Assessment and Risk Evaluation Process                                                      |
| AIGA                      | 022/13         | Code of Practice Acetylene                                                                                            |
| CGA                       | G-1<br>(2009)  | Acetylene                                                                                                             |
| ANSI/CGA                  | G-13<br>(2015) | Storage and Handling of Silane and Silane Mixtures                                                                    |
| ASME                      | B31.3          | Process piping                                                                                                        |
| EPA                       |                | <a href="#">List of Regulated Substances under the Risk Management Plan (RMP) Program</a>                             |
| Global EHS                | 08 Oct 2019    | <a href="#">Equipment Installation Standard</a><br>(2W4373RQWREN-1568922467-12)                                       |
| Global EHS                | 25 Dec 2019    | <a href="#">Toxic Gas Monitoring and Double Containment Standard</a><br>(2W4373RQWREN-1568922467-11)                  |
| Global PSM                | 16 Aug 2019    | <a href="#">Flammability Standard</a><br>(2W4373RQWREN-1568922467-212)                                                |

## 6.0 System Overview

### 6.1 Cylinder Spec Gas Supply Systems

- a. Gas Cabinet – An exhausted enclosure housing HPM specialty gas cylinder(s) and associated gas panel(s) for:
  - Connection to the cylinder(s)
  - Automatic shutoff and isolation of the cylinder(s)
  - Automatic operation including crossover from empty to standby cylinder
  - Cylinder pressure and/or weight monitoring and alarm
  - Delivery pressure monitoring, control, and alarm
  - Purge / Vent operations
  - Excess flow monitoring and control for HPM gases (as required by gas)
  - Filter
  - Gas Leak Detection Monitoring and Alarm
  - Enclosure Exhaust Monitoring and Alarm
  - Fire Detection Monitoring and Alarm
  - Fire Sprinkler Protection for flammable, toxic, highly toxic, pyrophoric, and oxidizer (excluding water reactive)
  - Connection to distribution piping or directly to the tool
  - Controller for automated operation, monitoring, alarming, and communications to SCADA
- b. Gas Rack – An open rack structure to support inert / non-HPM specialty gas cylinder(s) and associated gas panel(s) for:
  - Connection to the cylinder(s)
  - Automatic or manual operation including crossover from empty to standby cylinder
  - Cylinder pressure and/or weight monitoring and alarm
  - Delivery pressure monitoring, control, and alarm
  - Purge / Vent operations (optional)
  - Filtration / Purification (optional)
  - Connection to distribution piping
  - Controller for automated operation, monitoring, alarming, and communications to SCADA

### 6.2 Bulk Spec Gas Supply System

A supply system connected to a bulk source(s) (e.g., cylinder bundle, Y-cylinder, tonner, or tube trailer, etc.) of a specialty gas (HPM or inert), consisting of its associated gas panel(s) for:

- Connection to the bulk source(s)
- Automatic shutoff and isolation of the bulk source(s)
- Automatic operation including crossover from empty to standby bulk container
- Bulk Source pressure and/or weight monitoring and alarm
- Delivery pressure monitoring, control, and alarm
- Purge / Vent operations

- Excess flow monitoring and control
- Filter
- Gas Leak Detection Monitoring and Alarm for HPM
- Enclosure Exhaust Monitoring and Alarm for HPM
- Fire Detection Monitoring and Alarm for flammable, toxic, highly toxic, pyrophoric, and oxidizer
- Fire Sprinkler Protection for flammable, toxic, highly toxic, pyrophoric, and oxidizer (excluding water reactive)
- Connection to distribution piping
- Controller for automated operations, monitoring, alarming, and communications to SCADA.

### 6.3 Typical Equipment and Components

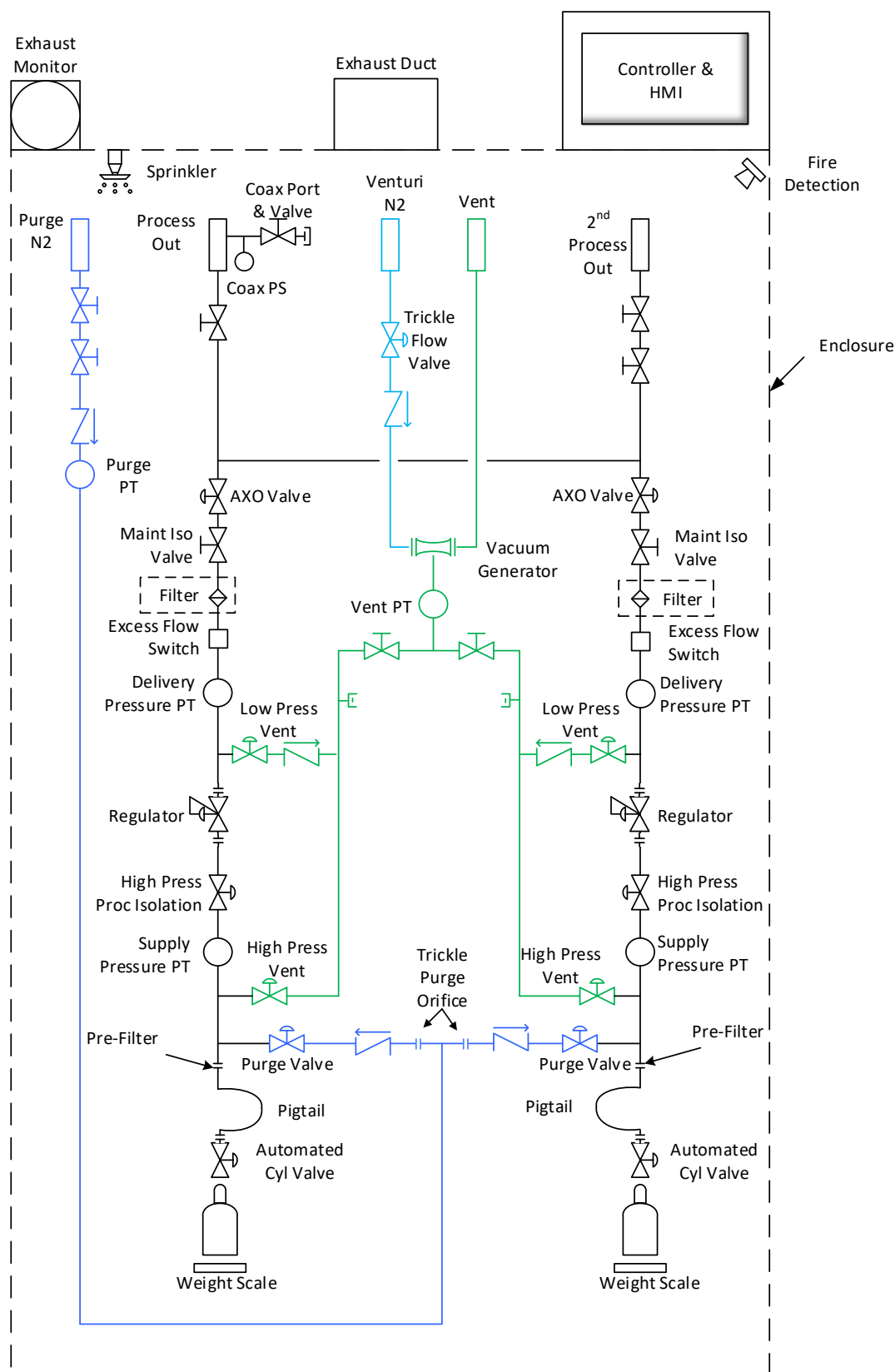
Refer to Figure 1 for typical process flow diagrams for supply systems. Equipment vendor PFDs may differ in configuration and component labelling.

- Pigtail – Rigid tubing including the cylinder connection fitting (CGA or DISS) for connecting the cylinder to the process panel.
- Automated Cylinder Valve – pneumatically actuated valve typically provided with HPM cylinders to provide for automatic shutdown of the cylinder in the event of a gas leak detection or other similar emergency event.
- Scale – to monitor the cylinder weight for liquefied gases (i.e., WF6, NH3, HF, etc.)
- Pre-filter – typically installed at VCR connection between pigtail and process panel to filter “large” particles to protect the regulator and downstream valves.
- Supply Pressure Transducer (PT) – provides for monitoring the cylinder pressure.
- High Pressure Process Isolation Valve – pneumatically actuated valve installed on the process panel upstream of the regulator. Isolates the regulator and low-pressure components from high pressure process gas, and provides for a smaller volume of tubing to cycle purge during the cylinder change process.
- Purge Valve – pneumatically actuated valve to provide for purge gas supply for the process panel in HPM gas cabinets. Purge valve may be provided as a manual valve for inert gas racks.
- Purge Pressure PT– provides for monitoring of the purge supply pressure.
- High Pressure Vent Valve – pneumatically actuated valve located on the gas panel upstream of the regulator to provide for evacuation of the process panel through the vacuum venturi in HPM gas cabinets. Vent valve may be provided as a manual valve for inert gas racks.
- AXO Valve – pneumatically actuated valve installed on the process panel downstream of the regulator to serve as the final gas shutoff valve and to isolate the panel from the downstream process during panel changeout. In auto-crossover cabinets,

this allows the switch from the empty cylinder to the standby cylinder.

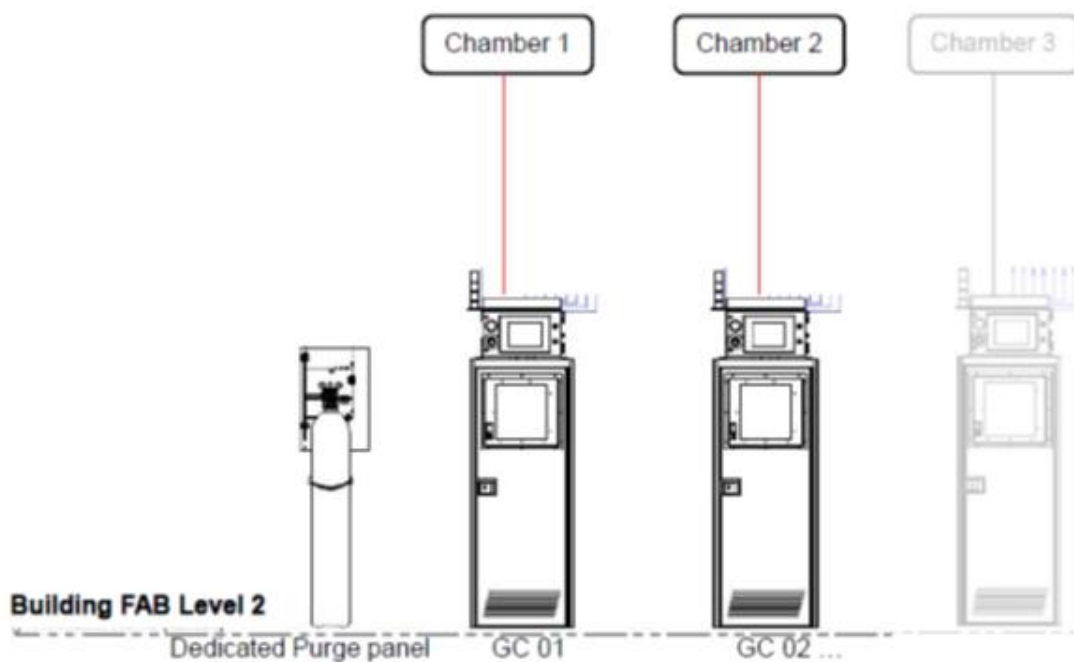
- Regulator – Regulator for adjusting the delivery pressure from the cylinder.
- Low Pressure Vent Valve – pneumatically actuated valve located on the gas panel downstream of the regulator to provide for evacuation of the process panel through the vacuum generator in HPM gas cabinets. May be provided as a manual valve for inert gas racks.
- Delivery Pressure Transducer (PT) – provides for monitoring the delivery pressure from the gas cabinet, gas rack, or BSGS.
- Excess Flow Switch/excess flow control (EFS) – provides protection against an excess flow condition by providing the signal for automatic shutdown.
- Maintenance Isolation Valve – manual valve installed on the outlet of the gas panel to provide for double valve isolation to perform maintenance on the gas panel in the event of a leak through the AXO valve.
- Vacuum Generator Module – an assembly consisting of a vacuum venturi, trickle purge orifice valve, and check valve to provide the source of vacuum used to evacuate the gas panel during cycle purges, and vents to the vent header.
- Vent Pressure Transducer (PT) – provides for monitoring the vacuum generated by the vacuum venturi.
- Filter – provides for final filtration (if required by modules).
- Coax Pressure Monitoring Assembly – an assembly of coaxial bulkhead fittings, valve, and pressure switch to provide for the means of evacuating (or pressurizing) and monitoring the annular space of coaxial tubing for those HPM gases that require secondary containment.
- Sprinkler – a 165°F (74°C) rated, quick response sprinkler head to provide fire protection in flammable, pyrophoric, and oxidizer gas cabinets and BSGS.
- Exhaust Duct – to provide ventilation and exhaust balancing for the exhausted enclosure for HPM gases.
- Exhaust Monitor – to provide for monitoring and alarm notification for the exhaust flow and differential pressure to ensure continuous exhaust ventilation through the exhausted enclosure.
- Fire Detection – to provide for detection and alarm notification for a fire within the exhausted enclosure for flammable, pyrophoric, and oxidizer gases. Fire Detection may be accomplished using a temperature switch, UVIR, or High Sensitivity Smoke Detector (HSSD).
- Controller (PLC or Microprocessor) – provides for automated operation of the gas cabinet, gas rack, or BSGS including alarm notifications, interlocks, and communications to SCADA. The controller includes the CPU, power supply, solenoids for pneumatic valves, I/O modules for the various digital and analog devices, and EMO (Emergency Manual Off) button.



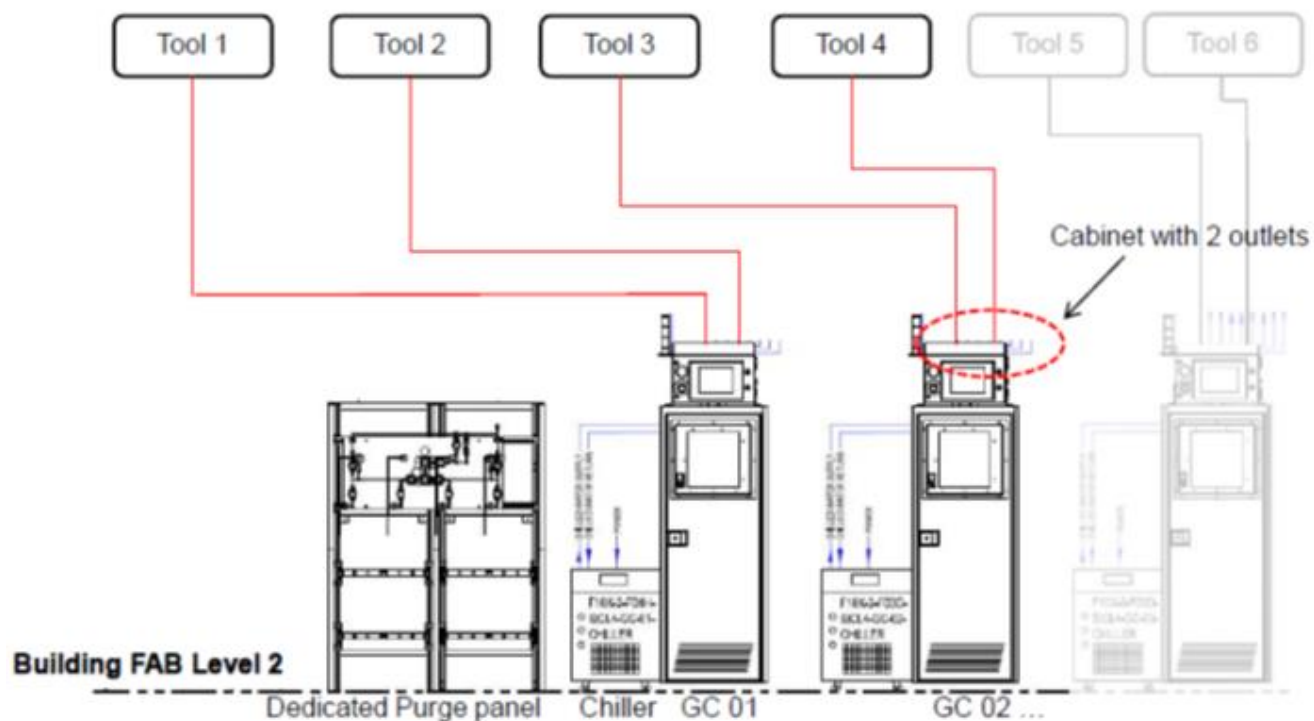
**Figure 1 – Typical PFD for Supply System**

## 6.4 Supply & Distribution System Configurations

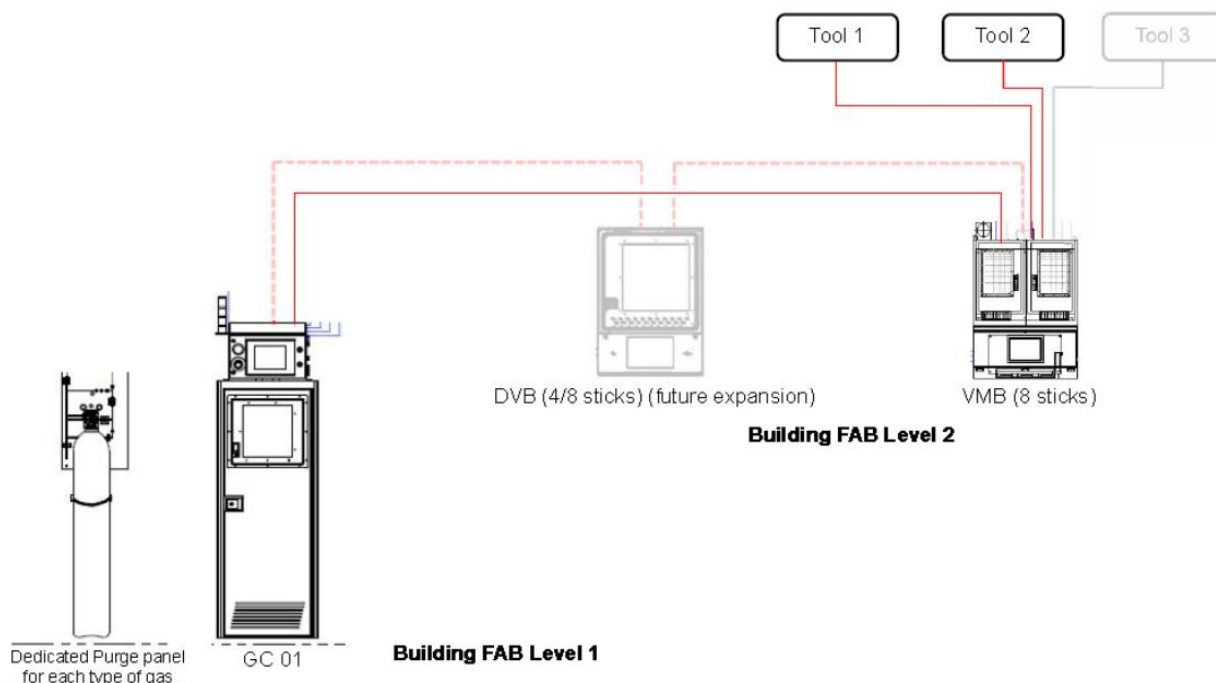
### a. Configuration 1 – One Gas Cabinet, One Chamber



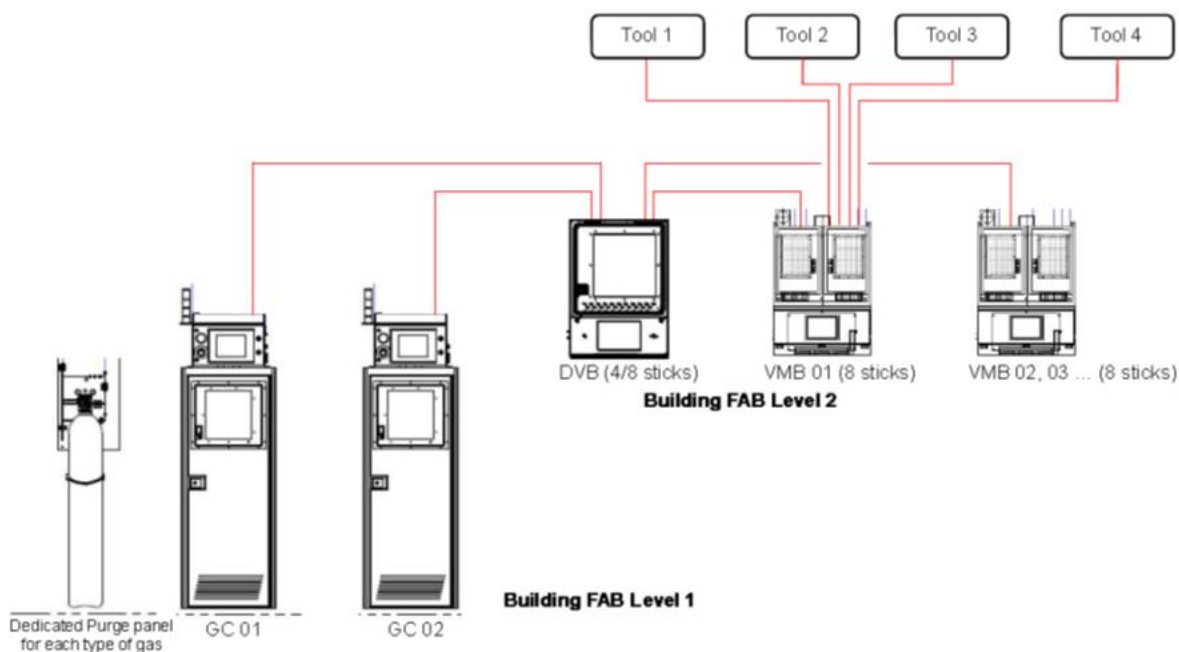
### b. Configuration 2 – One Gas Cabinet, Two Frames



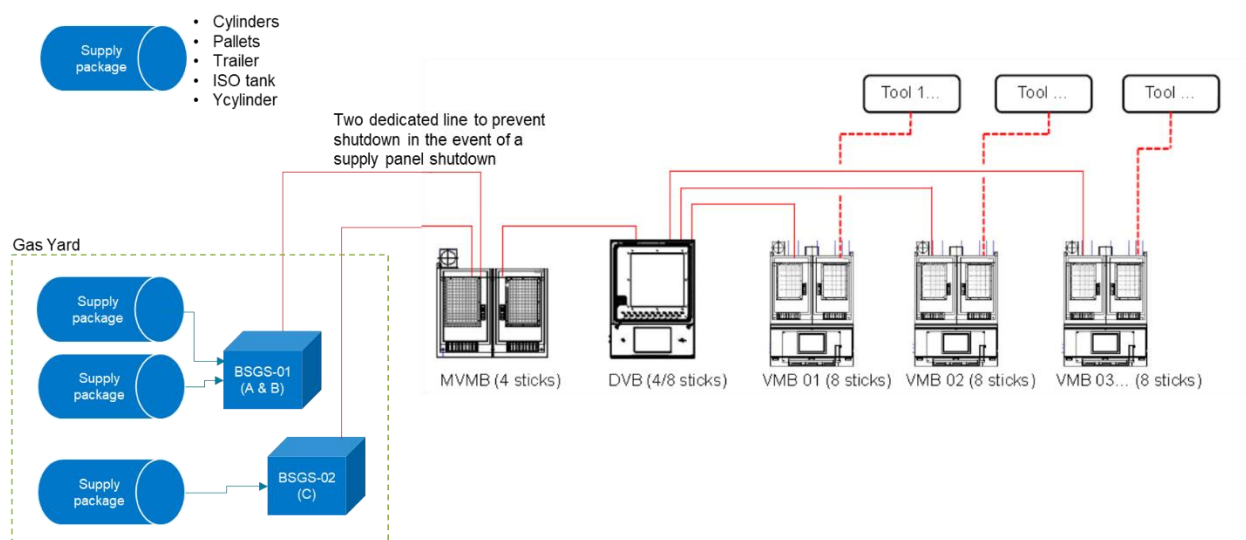
## c. Configuration 3 – One Gas Cabinet/Rack, One VMB/VMP



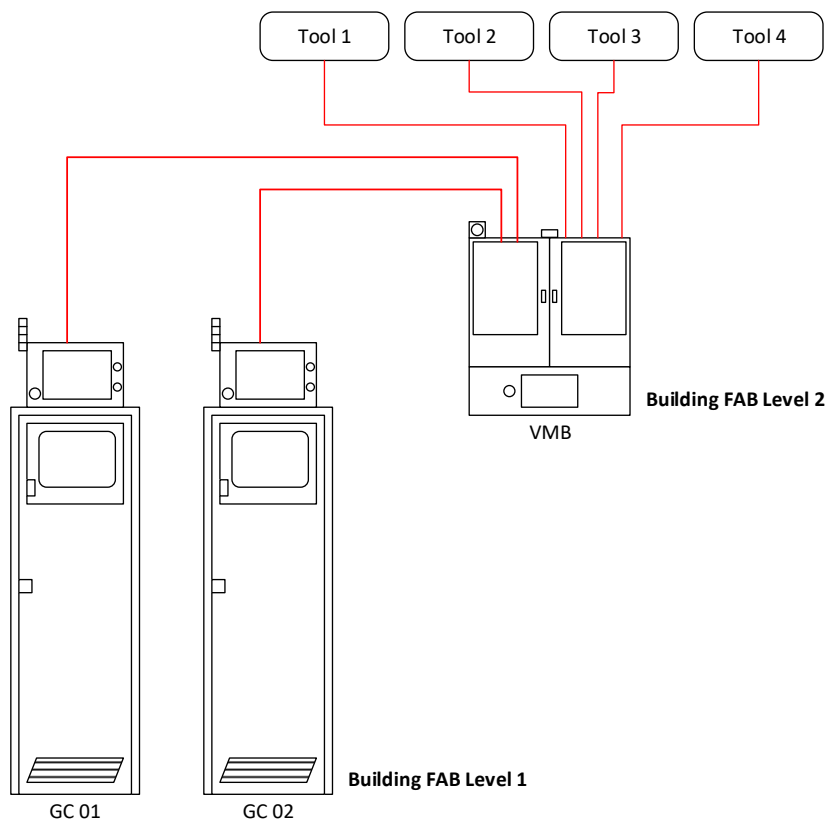
## d. Configuration 4 – Two Gas Cabinets/Racks, One DVB/DVP, Multiple VMB/VMP



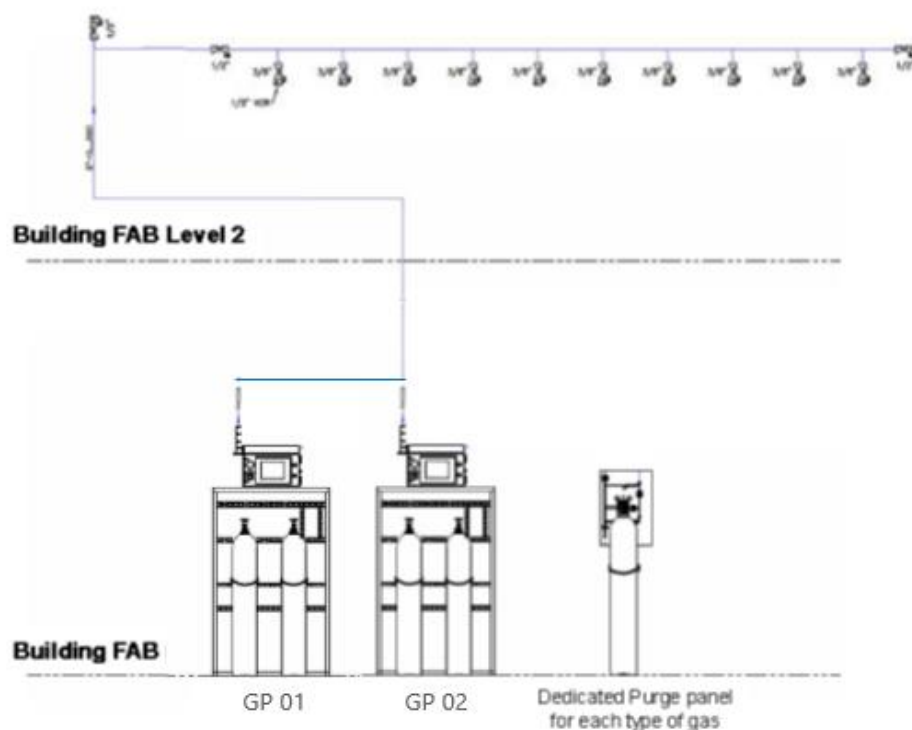
e. Configuration 5 – BSGS (N+1), One MVB, Multiple DVB, Multiple VMB



f. Configuration 6 – Two Gas Cabinets/Racks, One VMB/VMP



## g. Configuration 7 – One or two Gas Racks, Main/Laterals Distribution



## 7.0 HPM Room and Gas Room Requirements

## 7.1 Construction

- a. Separate Gas Rooms shall be provided for hazardous and flammable/pyrophoric gases, non-hazardous (including inert) and bulk gases, and grouped according to gas properties and safety requirements.
- b. Maximum Allowable Quantity per Control Area of Hazardous Material.
  - i. Consult the International Building Code for specific requirements related to Group H-2 and H-3 occupancies containing hazardous materials in excess of the quantities cited in IBC Section 307 "High-Hazard Group H". [IBC 307.1]
  - ii. Where the aggregate quantity of hazardous materials is in excess of those listed in IBC Section 307 Table 307.1(1) "Maximum Allowable Quantity per Control Area of Hazardous Materials Posing a Physical Hazard" and Table 307.1(2) "Maximum Allowable Quantity per Control Area of Hazardous Materials Posing a Health Hazard", the area shall be classified as Group H-5, and shall be constructed in accordance with IBC Section 415.11. [IBC 307.7]

- iii. The percentage of maximum allowable quantities of hazardous materials per control area permitted at each floor level within a building shall be in accordance with Table 414.2.2 within IBC section 414. [IBC 414.2.2]
    - iv. Follow local requirements if more stringent than guidance provided in IBC.
  - c. Floors & Ceilings
    - i. The floor shall have a fire-resistance rating of not less than 2 hours. [IBC 414.2.4]
    - ii. Protect openings or penetrations in floors, ceilings, and floor/ceiling assemblies such as utility and service openings with an FM approved fire stop assembly with a fire-resistance rating equal to or greater than that of the floor or ceiling. Use an FM Approved fire stop assembly that has a water leakage rating to protect floor penetrations. [FM Global Data Sheet 1-23, 2.1.2.3 and 2.1.2.4]
    - iii. Except for surfacing, floors of HPM rooms and gas rooms shall be of noncombustible liquid-tight construction. [IBC 415.11.5.3]
    - iv. Ceilings shall be coated with a primer and top coat. Follow site construction architectural specifications.
    - v. HPM room and gas room floor coatings shall be an approved chemical resistant coating (CRC). Follow site construction architectural specifications.
  - d. Walls
    - i. HPM Rooms and Gas Rooms shall have at least one exterior wall and such wall shall be not less than 30 ft from property lines, including property lines adjacent to public ways. [IBC 415.11.5.4]
    - ii. HPM rooms and gas room shall be separated from other areas by fire walls and the fire-resistance rating shall not be less than 2 hours where the area is 300 ft<sup>2</sup> (27.9 m<sup>2</sup>) or more and not less than 1 hour where the area is less than 300 ft<sup>2</sup> (27.9 m<sup>2</sup>). [IBC 415.11.5]
    - iii. Seal all penetrations in fire walls with an FM Approved wall penetration fire stop with a fire-resistance rating equal to or greater than that of the wall. Penetrations should not reduce the integrity of the wall. [FM Global Data Sheet 1-23, 2.1.1.4]
    - iv. Seal penetrations and protect openings in exterior walls designed for exposure protection. [FM Global Data Sheet 1-23, 2.1.1.6]
    - v. Walls shall be coated with a primer and top coat. Follow site construction architectural specifications.

- e. Doors
  - i. HPM rooms and gas rooms shall have two exits. One exit shall be directly to the outside of the building. [IBC 415.11.5.6]
  - ii. Doors in a fire barrier wall, including doors to corridors, shall be self-closing and have a fire protection rating of not less than  $\frac{3}{4}$  hour. [IBC 415.11.5.7]
  - iii. Ensure fire doors are FM Approved and have a minimum fire rating in accordance with Table 3 "Minimum Fire Door and Fire Shutter Ratings" in FM Global Data Sheet 1-23. [FM Global Data Sheet 1-23, 2.2.3.1]
  - iv. Ensure doors are self-closing (including normally closed doors). Initiate closure by either fire detection or sprinkler water flow. Locate the closing mechanism where it will be protected against mechanical damage. [FM Global Data Sheet 1-23, 2.2.3.3]
  - v. After installation, have fire doors satisfactorily tested, reset, and tested again to ensure the door-closing mechanism has been reset correctly. [FM Global Data Sheet 1-23, 2.2.3.24]
  - vi. Doors and doorways identified to be used for move-in of equipment and HPM cylinders/containers shall be sized to accommodate the largest anticipated package including transport mechanism (i.e., fork-lift, pallet jack, etc.) and installed with self-closing hardware.
- f. Explosion Control
  - i. Explosion control shall be provided where quantities of hazardous materials specified in IFC Table 911.1 exceed the maximum allowable quantities in IFC Table 5003.1.1(1), and such areas shall be provided with explosion (deflagration) venting, explosion (deflagration) prevention systems or barricades in accordance with IFC Section 911 and NFPA 69, or NFPA 495 as applicable. [IFC 911.1]
  - ii. Provide damage-limiting construction features in accordance with FM Global Data Sheet 1-44, Damage-Limiting Construction, when process gas cylinders containing pyrophoric (other than silane; see FM Global Data Sheet 7-108, Silane), flammable, or highly reactive compressed gases are located in the following locations [FM Global Data Sheet 7-7, 2.5.1.2.3]:
    - 1) Roof mounted room or cabinets
    - 2) Cabinets adjacent to outside of building
    - 3) Ground level cutoff room
    - 4) Second Story cutoff room

- g. Clearance between rows of gas cabinets
  - i. For greenfield construction, the clearance between rows of gas cabinets shall be not less than 48 in.
  - ii. For remodels of existing gas rooms, the clearance between rows of gas cabinets shall be not less than 36 in.

## 7.2 Emergency Alarms

- a. Emergency alarms for the detection and notification of an emergency condition shall be provided in Group H occupancies. [IBC 415.5]
- b. Provide a manual emergency alarm system for HPM rooms. An emergency alarm-initiating device shall be installed outside of the HPM room at the interior exit door to the corridor. Activation of the emergency alarm-initiating device shall sound a local alarm to alert occupants of an emergency situation involving hazardous materials. [IBC 415.5.1]
- c. Activation of an emergency alarm-initiating device shall transmit a signal to the emergency control station in the facility control room. An approved emergency telephone system, a local alarm manual pull station or other approved alarm-initiating devices are allowed to be used as emergency alarm-initiating devices. [IBC 415.11.5.9]
- d. The emergency alarm system shall be supervised by an approved central station in the facility control room. [IBC 415.5.3; Global Facilities – Design & Construction Standard – I&C – Fire Alarm and HSSD System 6.1]

## 7.3 Security

- a. HPM rooms shall be secured against unauthorized entry and safeguarded. [IFC 5303.5.1]
- b. Access into HPM rooms shall be controlled. Primary access into the HPM room shall be controlled by card reader. Secondary access into the HPM room from the outside may be controlled by lock and key.

## 7.4 Automatic Fire Detection System

- a. HPM rooms (Group H occupancies) shall be provided with an automatic fire detection system. [IBC 415.3]
- b. UVIR detectors shall be installed in flammable/pyrophoric rooms.

## 7.5 Automatic Sprinkler System

- a. HPM rooms (Group H occupancies) shall be equipped throughout with an automatic sprinkler system. [IBC 415.4]
- b. In HPM rooms (Group H occupancies), provide automatic sprinklers hydraulically designed to provide a density of 0.30 gpm/ft<sup>2</sup> (12 mm/min) over the entire room area or 3,000 ft<sup>2</sup> (280 m<sup>2</sup>), whichever is less. Use standard response sprinklers rated at 286°F (141°C). [FM Global Data Sheet 7-7, 2.3.4.7 B]



## 7.6 Electrical Design

- a. Electrical equipment and devices shall comply with NFPA 70. [IBC 415.11.1.8]
- b. Design rooms with grounding system for equipment
- c. Lighting shall provide 60 ft-candle coverage for normal working areas within the room. 60 ft- candle coverage shall be provided with fixtures designed not to obstruct access to overhead systems. LED lighting preferred for cost savings.
- d. All electrical components and wiring shall be listed. [NFPA 318, 8.4]
- e. Electrical enclosures should be suitable for the environment in which they are intended to be used. [SEMI S2, 13.4.6]
- f. Electrical equipment and devices within 1.5 m (5 ft) of gas cabinets in which flammable liquids or gases are used shall comply with the requirements of NFPA 70, for Class I, Division 2 locations. [NFPA 318, 8.4]
- g. Emergency Power [IFC 2703.15.1]  
Emergency power shall be provided for electrically operated equipment and connected control circuits for the following systems:
  - i. HPM Exhaust ventilation
  - ii. HPM gas cabinet ventilation systems
  - iii. HPM exhausted enclosure ventilation systems
  - iv. HPM gas room ventilation systems
  - v. Treatment system
  - vi. HPM Gas Detection system
  - vii. Temperature control system
  - viii. Emergency alarm systems
  - ix. Automatic sprinkler system monitoring and alarm systems

## 7.7 Room Ventilation

- a. Mechanical exhaust ventilation shall be provided in HPM rooms at a rate of not less than 1 ft<sup>3</sup>/min per ft<sup>2</sup> (0.044 L/s per m<sup>2</sup>) of floor area or six air changes per hour. Exhaust ventilation for HPM and gas rooms shall be designed to operate at negative pressure in relation to surrounding areas and direct the exhaust ventilation to an exhaust system. [IBC 415.11.5.8]
- b. Exhaust ducts shall not penetrate fire walls. [IBC 414.11.1.6]
- c. Fire dampers shall not be installed in exhaust ducts. [IBC 414.11.1.6]
- d. Mechanical air handling equipment shall not be located within HPM rooms and gas rooms

- e. A manual ventilation system shutoff shall be provided outside of the room in a position adjacent to the access door to the room or in an approved location. The switch shall be a break-glass or other approved type and shall be labeled: VENTILATION SYSTEM EMERGENCY SHUTOFF. [IFC 5004.3.1]
- f. Exhaust air shall not be recirculated to occupied areas if the materials stored are capable of emitting hazardous vapors and contaminants have not been removed. [IFC 5004.3.1]

## 7.8 Room Environment

- a. Environmental control shall be maintained at  $70 \pm 2$  °F.
- b. A minimum of two temperature monitors shall be provided in HPM rooms and gas rooms with temperature averaging and a local temperature display mounted inside the room; this shall be collected and stored in SCADA.
- c. To help maintain the room temperature, a door alarm on all exterior doors to the outside shall be installed. It will indicate that the door is in the open position.

## 7.9 Life Safety Systems

- a. Safety showers and eye wash stations shall be placed in each HPM-Gas Rooms and Gas Utility Rooms. Safety showers and eyewash stations shall be installed in accordance with all applicable codes and regulations.
- b. All safety shower and eyewash stations will be installed with a drain, activation monitoring, and alarm back to the control room in areas that are not typically occupied spaces and have significant safety and health hazards.
- c. A life and safety drawing shall be provided to include the occupancy rating and locations of all life and safety systems within each Gas Room and each Gas Utility Room.
- d. A gas detection system shall be provided in HPM rooms where HPM gas is used in the room. [IFC: 2703.13.1.2]
- e. HPM gas systems shall be equipped with an appropriate Toxic Gas Monitoring (TGM) System, according to the Global EHS - [Toxic Gas Monitoring and Double Containment Standard](#)
- f. Emergency alarms (horns and strobes) shall be provided HPM gas rooms. [IBC 415.5]
- g. Emergency alarm-initiating devices shall be installed outside of each interior exit or exit access door of HPM rooms. Activation of an emergency alarm-initiating device shall sound a local alarm to alert occupants of an emergency situation involving hazardous materials. [IBC 415.5.1]
- h. UVIR detectors shall be installed in flammable/pyrophoric rooms and pyrophoric supply systems.
- i. Oxygen monitoring to be provided for inert gas supply area and possible confined space where inert services is in vicinity

### 7.10 Room Layout

- a. All gas cabinets used for gas source delivery will be located in a gas room properly designed to contain the appropriate gas type and hazard rating. Exception: Extremely low vapor pressure gases (i.e., WF6) which are located in the subfab directly beneath the tool it serves.
- b. To allow access and serviceability of equipment, minimum aisle spacing between rows of cabinets or panels shall be not less than 48 inches for new greenfield construction and 36 inches for remodels of existing gas rooms.
- c. Gas cabinet and gas panel installation within all gas rooms must be designed so that any single cabinet or panel can be installed or removed without impacting other gas equipment.
- d. Gas cabinets containing flammable materials shall have a grounding screw for attaching to the grounding grid.
- e. Gas cabinets shall be securely anchored to the floor for seismic considerations.

### 7.11 Pipe/Tubing/Exhaust/Conduit Routing

- a. All gas room utility piping, exhaust ducts, and wire conduit routing shall be designed to be distributed within the room in the area above gas cabinet connection zones to provide access to facilitate and modify gas line changes.
- b. Gas distribution piping will be designed to be routed above/away from foot traffic and clear of all mechanical devices that would cause vibration or physical damage to the piping or mounting systems.
- c. Piping runs will be located within a clear path and located to minimize length, drops, rises, and bends.
- d. Piping runs shall be designed to maintain specified temperature (70 +/- 2 °F) from the source to the point of use connection.
- e. Route gas distribution piping to allow access for future installations of gas lines and install piping with clearance for two lines per each gas panel or gas cabinet plus a 20% spare capacity.
- f. Route gas distribution piping in sequential order on vertical racking above supply gas panels or gas cabinets. Vertical racking will be utilized for all horizontal piping runs if possible.
- g. Route gas distribution piping through walls utilizing bulkhead feed through plates designed to allow continuous piping runs and meet or exceed fire-resistance rating of the wall as applicable.
- h. Gas distribution sub-systems and equipment shall be designed for locations in accessible areas that are clear of powered equipment or materials movement routes.
- i. Gas rooms shall be constructed with vertical system specific zoning. Each zone shall be designed to contain only those systems that have been authorized to occupy that space.
- j. Do not install liquids and water services piping through the room except for fire sprinkler.

## 7.12 Operational Considerations

- a. HPM and gas rooms shall have a cylinder staging area provided with hardware for cylinder restraints, clear delivery path, and clear exit path.
- b. HPM and gas rooms shall have access to a shipping/receiving dock. Where hazardous materials having a ranking of 3 or 4 in accordance with NFPA 70 are transported through corridors or exit passageways, there shall be an emergency telephone system, a local manual alarm station or an approved alarm-initiating device at not more than 150 ft intervals and at each exit and exit access doorway throughout the transport route. The signal shall be relayed to the Emergency Alarm central station in the facility control room and shall initiate a local audible alarm. [IBC 415.5.2]
- c. All gas rooms shall be located as close to the point of use as possible to minimize length of piping runs from source to point of use.
- d. HPM rooms containing water-reactive materials shall be resistant to water penetration. Do not install liquids and water services piping through the room except for fire sprinkler.
- e. Gas Utility Room / Area
  - i. A Gas Utility Room may be provided to support all Non-HPM and HPM Gas Rooms. A Gas Utility Room may be provided for gas detection systems, automatic gas shutdown systems, wall-mounted self-contained breathing apparatus, cabinet storage for local supplies and PPE equipment, local communications, and/or electrical equipment.
  - ii. Gas Utility Room shall have an exit to the outside of the building that is independent of the HPM or gas rooms that it is serving.
  - iii. Gas Utility Room shall have or be adjacent to a shipping/receiving dock with a clear transport path from dock to adjacent gas distribution rooms. Area to be used for gas cylinder deliveries.
  - iv. Gas Utility Rooms may need to meet occupancy requirements based on location and HPM material transport path.

## 8.0 Specialty Gas Supply Systems

### 8.1 Design Considerations

- a. The project team involved in the design phase shall include team members from the Operations & Maintenance, and Safety identified at each site.
- b. Main design parameters include, but are not limited to the following:
  - i. POC Delivery Pressure – the required pressure at the tool point of connection.
  - ii. Peak Flow Demand – the flow demand when all tools connected to the system require gas simultaneously.
  - iii. Average Flow Demand – the average flow of the system on a daily, weekly, and monthly basis according to the gas usage in the same site or other sites, comparable for technology and manufacturing device. For FOAK (first-of-a-kind) gas, the TSR shall be used as a reference.
  - iv. Design Flow – the sustainable flow rate as provided by the equipment supplier.
  - v. Redundancy – the need for additional supply systems in an N+1 configuration, assuming uninterrupted supply to the fab.
  - vi. Toxic Gas Monitoring Requirements – based upon Global EHS Toxic and Flammable Gas Monitoring Requirements.
  - vii. Tool Quantity – the total number of tools to be supplied by the system.
  - viii. Manufacturing area sharing/loading – the sharing of a system to different production areas to restrict the potential impacts in the event of a system shutdown and/or contamination.
  - ix. Gas package size – the amount of gas contained in the available cylinder. As a general rule, the package will be identified with its biggest size, except for those gases that may have specific limitation and restriction by local codes and regulations, and/or EHS limitation.
  - x. Package Changeout Frequency – For gas cabinets, the cylinder changeout frequency should be not less than every 3 days. For BSGS, the bulk package (Y-cyl, tonner, or tube trailer) changeout frequency should be not less than every 7 days.

## 8.2 Gas Cabinet & BSGS Enclosure

- a. Enclosures shall be of noncombustible construction. [IFC 5003.8.5.1]
- b. Enclosures shall be constructed with the following [IFC 5003.8.6.1]:
  - i. Not less than 0.097 in (2.5 mm) (No. 12 gage) steel.
  - ii. Self-closing limited access ports or noncombustible windows to provide access to equipment controls.
  - iii. Self-closing doors.
  - iv. Interiors treated, coated or constructed of materials that are compatible with the hazardous materials stored.
- c. Gas cabinets shall be configured for 2 cylinders.
- d. Equipment enclosures should be designed so that if it is anchored as specified in the OEM documentation, and it experiences anticipated seismic forces, it will not overturn, and no parts will fail or yield such that there would be a risk of injury to personnel, or adverse environmental impact. [SEMI S2, 19.1]
- e. All enclosures shall be provided with a means of connection to the electrical ground grid.

## 8.3 Gas Cabinet & Enclosure Exhaust

- a. Gas cabinets shall be provided with an exhaust ventilation system. The ventilation system shall be designed to operate at a negative pressure in relation to the surrounding area. [IFC: 5003.8.6.2]
- b. Gas cabinets containing toxic or highly toxic compressed gases shall also meet the following [IFC: 6004.1.2]:
  - i. The average ventilation velocity at the face of the gas cabinet access ports or windows shall be not less than 200 ft/min (1.02 m/s) with not less than 150 ft/min (0.76 m/s) at any point of the access port or window.
  - ii. Gas cabinets shall be connected to an exhaust system
  - iii. Gas cabinets shall not be used as the sole means of exhaust for any room or area.
- c. Gas cabinet exhaust shall be monitored and alarm to the SCADA in case of an exhaust system failure. [NFPA 318: 6.4.4]
- d. The exhaust for gas cabinets and exhausted enclosures containing corrosive, toxic, and highly toxic gases will be directed to a treatment system designed to process the accidental release of gas. [NFPA 55: 7.5.3.4; 7.9.3]
- e. Provide a point-of-use (POU) gas treatment/abatement system if flammable gases in concentrations above 25% of their LEL can be present in purge panel vent lines, process equipment purge lines,

or process equipment vacuum pump exhaust. [FM Global Data Sheet 17-12: 2.3.5.9.2.2]

f. Emergency Exhaust Requirement for Taiwan

In Taiwan, emergency exhaust scrubber is required for the following gases:

- i. ClF<sub>3</sub>
- ii. 20%F<sub>2</sub>/N<sub>2</sub>
- iii. F<sub>2</sub>/Ar/Ne
- iv. F<sub>2</sub>/Kr/Ne
- v. Cl<sub>2</sub>
- vi. 1%PH<sub>3</sub>/He
- vii. 5%PH<sub>3</sub>/N<sub>2</sub>
- viii. 0.5%PH<sub>3</sub>/N<sub>2</sub>

#### 8.4 Venturi Vent & Vent Headers

- a. The release of purged gases or mixtures containing any quantity of corrosive, toxic, or highly toxic gases shall be through a treatment system. [NFPA 55, 7.1.18.3.2.1]
- b. Vent Headers [NFPA 318, 7.1.5]
  - i. Purge panel vent line headers shall be designed to prevent the mixing of incompatible gases and pyrophoric gases with air.
  - ii. ~~Silane~~ All vent headers or individual purge panel vent lines shall have a continuous flow of purge nitrogen to prevent back diffusion of air into the vent line by introducing the purge nitrogen upstream of the first exhaust connection to the header. The purge nitrogen gas velocity in the vent line shall be no less than 1 ft/s (0.3 m/s).
  - iii. The purge nitrogen flow into the vent header will be monitored and provide an alarm when the flow falls below 1 ft/s. (
  - iv. Vents shall terminate at a safe location or in treatment systems.
- c. Connect gas panel venturi vent lines for the same family of compatible gases into a common header/manifold and direct to appropriate treatment system. No dampers should be placed on the gas cabinet panel vents or exhaust duct that is 2 inches or smaller.
- d. Gas cabinet vents shall be connected to the vent manifold headers at the top of the tubing. Vent headers shall be a minimum of 1.5" diameter tubing to the abatement unit or to the Teflon-lined section of the scrubber exhaust system. The number of gas cabinet or VMB vent connections to a single header shall be limited by the abatement system capacity. Verify capacity of vent header before adding vent loads so as not to overload the abatement system.

- e. High pressure valves shall be used for vent valves in high pressure gas service.
- f. Vent lines for Silane and Silane mixtures
  - i. Vent lines for Silane, silane mixtures, and gases containing silane with an inert purge gas shall be dedicated and not be piped into the same vent header with other incompatible gases. [CGA G-13, 10.1.4.3, 14.5]
  - ii. Vent lines for silane and silane mixtures are to be purged continuously with an inert gas and directed to a treatment system. The inert purge gas velocity in the vent line shall be no less than 1 ft/s (0.3 m/s). [CGA G-13, 14.5.1]
  - iii. Nitrogen shall be used to continuously purge the vent line to atmosphere or otherwise arranged to preclude either the backflow of air or the development of backpressure at the point of discharge. A purge gas flow of no less than 1 ft/s (0.3 m/s) is required. [CGA G-13, 14.5.3]
  - iv. Silane regulator bonnet relief vents are to be vented to a safe location in the event of a diaphragm leak or rupture, and sized so as not to restrict the flow from the bonnet relief vent. [CGA G-13: 10.2.2.3]

#### 8.5 Purge Panel [NFPA 318: 7.1.4]

- a. Purge panels shall be provided at the cylinders for all HPM gases.
- b. Purge panels shall be designed to prevent backflow and cross-contamination of purge gas or other process gases.
- c. A manual isolation valve shall be provided on the process delivery line at the purge panel to permit removal of the purge panel for repair and maintenance.
- d. A dedicated purge gas system may only supply purge N2 to HPM process gas panels of compatible hazard categories. Corrosive and oxidizer gases may share a common purge system, and flammable and pyrophoric gases may share the same purge system provided there are no incompatibilities within these groups of gases. Purge gas from house N2 shall not be permitted.
- e. A dedicated purge gas system should consist of N2 gas cylinder(s), a purge panel, and purge header with POC valves for connection to the purge panels within the individual gas cabinets.
- f. Purge gas for silane delivery systems shall be supplied from dedicated inert gas supply source. A purge gas manifold shall be provided when one inert purge gas cylinder or supply is used as the common purge gas source for multiple silane systems. House gas supplies shall not be used for purging silane delivery systems at the source location due to the risk of backflow into the house gas supply. [CGA G-13: 15.2; 15.2.1]



## 8.6 Process Panel

- a. Gas supply systems shall be equipped with auto-crossover manifolds connected to a gas package to guarantee continuous supply. Typical configuration is one side supplying gas and the other side is in standby, ready to supply.
- b. Each process panel shall be equipped with a pressure regulator to reduce pressure from the full package pressure. Tied diaphragm regulator shall be used in gas supply systems to < 100 psig, except for specific applications where a higher delivery pressure may be required. Both high and low pressure sides (pre/post regulator) shall be equipped with connections to venturi vent with isolation valves.  
Exception: Regulators shall not be installed in HF gas cabinets supplying 21VE tools. See 8.19.g HF for 21VE Tools.
- c. Where HPM gases are delivered at pressures above 15 psig, excess flow control (typically an excess flow switch) shall be provided for emergency shutoff.
- d. Process panels shall be provided with a purge gas connection on the high pressure side (upstream of the regulator) of the panel with double valve isolation.
- e. Double isolation with an air-operated valve (AOV) and a manual valve shall be provided on the low pressure side of the panel to isolate the two sides.
- f. Dual process outlet with double valve isolation shall be provided in gas supply systems.
- g. Bypass mechanisms that are provided to manually force open AOV's to restore gas flow in the event of a controller failure must have a method to be operationally locked out to prevent unauthorized or unintentional use. Operation of the bypass mechanism and restoration to normal operating condition shall be controlled by procedure.
- h. Final Filters

| <b>Gas Type</b>                      | <b>Filter Media</b> | <b>Filter Size (µm)</b> | <b>Filter Body</b>           |
|--------------------------------------|---------------------|-------------------------|------------------------------|
| Inert                                | Stainless Steel     | 0.003                   | 316L SS, EP, < 10 Ra         |
| Flammable                            | Stainless Steel     | 0.003                   | 316L SS, EP, < 10 Ra         |
| Corrosive                            | Stainless Steel     | 0.003                   | 316L SS, VIM-VAR, EP, < 5 Ra |
| Strong Corrosive & Oxidizer (Note 1) | Hastelloy           | 0.003                   | 316L SS, VIM-VAR, EP, < 5 Ra |

Note 1: N2O, NF3, NO, Cl2, BCl3, ClF3, HBr, BF3, 20% F2/N2, COS

- m. Gasket material for cylinder CGA connections and VCR fittings
  - i. Stainless steel (SUS) gaskets shall be used for CO, C<sub>2</sub>H<sub>2</sub>, COS, ASH<sub>3</sub>, PH<sub>3</sub>, B<sub>2</sub>H<sub>6</sub>, H<sub>2</sub>Se. Nickel and Ni-plated gaskets shall not be used for these gases as they are not compatible with Ni.
  - ii. For all other gases, Nickel and Ni-plated gaskets are acceptable for use.

- n. Tubing

All material and components shall be rated for the maximum pressure allowable in the gas package with 20% over-capacity as minimum. Depending on specific application, 316L SS EP (electro-polished), tubing as minimum, rated in the following range of operating pressures shall be provided:

| Tube OD | Wall Thickness (inch) | Max. Allowable Working Pressure (psig) |
|---------|-----------------------|----------------------------------------|
| 1/4"    | 0.035                 | 5100                                   |
| 3/8"    | 0.035                 | 3300                                   |
| 1/2"    | 0.049                 | 3700                                   |
| 3/4"    | 0.065                 | 3300                                   |
| 1"      | 0.065                 | 2400                                   |

- o. Valves

- i. For inert and flammable gases, use 316L SS EP, <10 Ra
- ii. For Corrosive gases, use 316L SS EP, VIM-VAR, < 5 Ra
- iii. Valves for N<sub>2</sub>O service shall have Vespel seats.
- iv. All valves shall be rated with appropriate flow capacity, as indicated by manufacturers with the component "Cv", able to meet or exceed the required capacity.
- v. For B<sub>2</sub>H<sub>6</sub>, HF, BF<sub>3</sub>, Cl<sub>2</sub>, ClF<sub>3</sub>, HCl, HBr, DCS, 20%F<sub>2</sub>/N<sub>2</sub>, and SiCl<sub>4</sub>, the "low pressure" valves located downstream of the regulator shall be the same model of "high pressure" valves located upstream of the regulator. For these gases, all in-line process valves are to be the same.
- vi. Manual valves shall be provided with an integral method for locking the valve in the closed position, and providing visual indication of OPEN or CLOSE position. Applicable to all manual valves within the gas supply system.
- vii. Process gas pneumatic valves shall be provided with a Rotowink pressure indicator to indicate the valve's OPEN or CLOSE position.
- viii. For pneumatic valves, PFA (perfluoroalkoxy "Teflon") is the preferred material for pneumatic tubing. If PU (polyurethane) is used for pneumatic tubing, a preventive maintenance plan shall be implemented for inspection and periodic replacement as directed in FCT Operations and Maintenance BKM Standard. Applicable to all AOV valves within the gas supply system.

ix. Vent valves in high pressure gas service shall be the same model of "high pressure" valves located upstream of the regulator.

p. Regulator

- i. For inert and flammable gases, use 316L SS EP, <10 Ra
- ii. For corrosive gases, use 316L SS EP, VIM-VAR, < 5 Ra
- iii. For B2H6, use AP Tech AP1710SM series
- iv. For N2O and CO2, regulator seat material shall be Vespel
- v. For liquefied low-pressure gases (WF6, C5F8, SiCl4, HF, ClF3, DCS, etc.) use an absolute pressure regulator.
- vi. For BSGS packages (cylinder pack, Y-Cyl, Tube Trailer, etc.) use "high flow" series regulator with maximum outlet pressure  $\geq 130$  psig.

#### 8.7 Automatic Cylinder Connection Protection Device ("Auto-Guard")

- a. BSGS containing HPM gases shall be provided with an automatic cylinder connection protection device which prevents the accidental disconnection of the hazardous gas cylinder from the pigtail.
- b. The device shall extend a shroud or cover over the cylinder connection to prevent access to the cylinder connection and to indicate that it is unsafe to disconnect the cylinder.
- c. A retracted shroud allows access to the cylinder connection and indicates that it is safe to disconnect the cylinder.
- d. The device shall be automatically operated by the gas supply system controller.

#### 8.8 Gas Detection

- a. A continuous gas detection system must be provided for HPM gases when the physiological warning properties of the gas are at a higher level than the accepted Occupational Exposure Limit for the gas, and for flammable gases. [[Global EHS – Toxic Gas Monitoring and Double Containment Standard](#): 6.1.2]
- b. Monitoring points are required where the HPM gas is used or stored in gas cabinets, exhausted enclosures, and gas rooms. [[Global EHS – Toxic Gas Monitoring and Double Containment Standard](#): 6.1.6]
- c. The continuous gas detection system shall be capable of monitoring the gas at or below the following gas concentrations: [IBC: 415.11.7.2]
  - i. IDLH values where the monitoring point is within an exhausted enclosure or gas cabinet.
  - ii. For flammable gases, vapor concentrations in excess of 25% LFL where the monitoring point is within an exhausted enclosure or gas cabinet.

- d. Where technically feasible, the gas detection system should be able to detect a toxic gas at a concentration at least 20% of the TLV-TWA, a flammable gas at a concentration at least 5% of the LFL. [[Global EHS – Toxic Gas Monitoring and Double Containment Standard](#): 6.1.4]
- e. The gas detection system shall initiate a local alarm and transmit a signal to the central control room that is responsible for notifying and dispatching ERT teams. The alarm shall be both visual and audible and shall provide warning both inside and outside the area where the gas is detected. [IBC: 415.11.7.2.1; [Global EHS – Toxic Gas Monitoring and Double Containment Standard](#): 6.1.9]
- f. Refer to Global EHS – Toxic Gas Monitoring and Double Containment Standard for Alarm and Shutoff levels for toxic and flammable gases.
- g. Gas detection points should be placed in the exhaust of the gas cabinets. In addition, one general room air monitoring point should be placed in each gas room. [[Global EHS – Toxic Gas Monitoring and Double Containment Standard](#): 6.1.6]
- h. The gas detection system shall automatically close the shutoff valve at the source for the specific gas detected within the gas cabinet. [IBC: 415.11.7.2.2]
- i. The gas detection system shall provide continuous monitoring of gas leaks during normal operation and following a loss of power. The gas detection system shall be supplied with a reliable supply of power to ensure that it can continue to monitor for gas leaks for 24 hours plus a further 5 minutes in alarm mode, either through the use of a battery backup or connection to a reliable emergency power supply. [NFPA 318: 7.1.6.3.1]

## 8.9 Seismic Protection

- a. In seismic zones, a seismically activated valve should be provided to activate emergency shutoff valves on gas cabinets and BSGS. [NFPA 318: Annex B.1.2]
- b. Gas cabinets and enclosures should be designed so that if they are anchored as specified in the OEM documentation provided to the equipment user, and it experiences anticipated seismic forces, it will not overturn, and no parts will fail or yield such that there would be a risk of injury to personnel, or adverse environmental impact. [SEMI S2: 19.1]
- c. Refer to SEMI S2, Section 19 "Seismic Protection" for design requirements and calculation of the anticipated horizontal seismic forces to be withstood.

## 8.10 Fire Protection

- a. Sprinklers shall be installed in gas cabinets that contain flammable, toxic and highly toxic, oxidizer, and pyrophoric gases (including silane). It is recommended that a listed 57°C (135°F), 9.5 mm (3/8 in.) orifice sprinkler be installed. [NFPA 318: 11.2.4]
- b. EXCEPTION: Sprinklers are not required in gas cabinets containing Class 3 Water Reactive gases. [NFPA 318: 11.2.4]
- c. Connection from the sprinkler lateral to a gas cabinet shall use a flexible head style connection.
- d. Gas cabinets containing toxic or highly toxic gases located in the Fab level shall be equipped with automatic sprinkler protection. [IFC: 2704.2.1; 6004.1.2]
- e. Silane Flame Detection and Fire Protection
  - i. When silane delivery systems are located outdoors, an optical flame detection system shall be located to detect fires at the connection to the source containers (i.e., cylinder racks, cylinder packs, ISO modules, etc.), process panels, and purge panels. [CGA G-13: 11.2.1]
  - ii. When silane delivery systems and silane cylinder staging areas are located indoors, an optical flame detection system shall be provided within the room. [CGA G-13: 11.2.2].
  - iii. Optical flame detection system shall be provided inside of silane gas cabinets and associated purge systems. [CGA G-13: 11.2.2.2]
  - iv. A manually activated deluge water spray fire protection system shall be provided to protect bulk silane BSGS. In the event of fire, silane bulk sources (Y-cylinders, Tube Trailers, etc.) shall be cooled if the source cannot be shut off. Automatic deluge systems are allowed as an option to manual activation. The deluge system shall provide a minimum density of 0.3 gpm/ft<sup>2</sup> (12 lpm/m<sup>2</sup>) for a minimum of a 2 hr duration over the external container surface area including the silane containers. [CGA G-13: 12.2.1 – 12.2.1.3]

## 8.11 Excess Flow Control

Where gases having a hazard ranking of Health Class 3 or 4, Flammability Class 4 or Instability Class 3 or 4 are carried in pressurized piping above 15 psig (103 kPa), excess flow control shall be provided. [IFC: 5005.1.12]

## 8.12 Cylinder Heating & Heat Trace Systems [NFPA 318: 8.5]I

- a. The electrical supply shall include a ground-fault circuit interrupt breaker with overcurrent protection when package/cylinder is heated by means of a resistive heater.
- b. Automatic temperature process control shall be included.
- c. A power interrupt circuit with manual reset that removes power to the heating elements when activated shall be included.
- d. Process overtemperature interlock protection shall be provided.
- e. Heater overtemperature interlock protection shall be provided.
- f. Heater controllers shall be provided with dual power supplies.

## 8.13 Emergency Isolation / Automatic Shutdown

- a. Emergency isolation shall be provided for compressed gases having a health hazard Class 3 or Class 4, Flammability Class 4, or Instability/Reactivity Class 3 or Class 4, and are carried in pressurized piping above 15 psig (103 kPa). [NFPA 55: 7.3.1.12.1; IFC: 5005.1.12]
- b. Acceptable means of emergency isolation include: [NFPA 55: 7.3.1.12.2]
  - i. Automatic shutoff valves at the source or as close to the source as practical, tied to leak detection systems.
  - ii. Excess flow valves at the source.
- c. For HPM gas cylinders with a manual cylinder valve, automatic shutdown may be provided by a "Shut-boy" installed on the manual cylinder valve, or the nearest automatic valve to the cylinder installed on the pigtail or panel.
- d. For all gases, provide remote manual actuation of the automated cylinder valves and emergency shutoff valves outside the gas room. [FM Global Data Sheet 17-12: 2.5.1.4.7]
- e. Interlocks shall shutdown the supply of gas when an unsafe condition is detected or initiated:
  - i. Gas Leak Detection: Design shall include a leak detection system for hydrides, mineral acids, oxidizers, and flammables, as specified in the Global EHS [Toxic Gas Monitoring and Double Containment Standard](#). Shutdown occurs at the first available equipment upstream from the point of leak detection.
  - ii. Excess Flow
  - iii. High Pressure
  - iv. Emergency Shut Off (ESO) / Emergency Manual Off (EMO) Button
  - v. Fire (UVIR/Temperature Switch/Rate of Rise/High Sensitivity Smoke Detector)

## 8.14 Secondary (Double) Containment

- a. Micron requires secondary containment piping for Class I (highly-toxic) and Class II (toxic) gases where the gas being conveyed is corrosive to the primary piping. [Global EHS – [Toxic Gas Monitoring and Double Containment Standard](#): 6.2.1]
- b. Refer to the Double Containment Matrix which specifies the double containment requirement for individual HPM gases. [Global EHS – [Toxic Gas Monitoring and Double Containment Standard](#) : 6.2.1]
- c. Inner and outer containment lines must be continuously welded and terminate within an exhausted enclosure (gas cabinet or valve manifold box).
- d. Inner tubing shall be no less than 316L SS EP, 5 Ra, seamless tubing.
- e. Outer tubing shall be no less than 304L SS BA CFOS tubing.
- f. Mechanical fittings are not allowed except within exhausted and monitored enclosures.
- g. Closed secondary containment systems shall have the same leak integrity as the primary containment except the closed secondary containment shall be leak tested with the annulus pressurized to 2 times the maximum operating pressure with argon containing at least 10% helium. There should be no drop in pressure for a period of 24 hours.
- h. For non-liquefied gases at a pressure > 20 psig, the annular space will be maintained at a vacuum and vacuum decay shall be used to monitor the annular space and alarm/interlock on loss of vacuum.
- i. For liquefied gases at a pressure > 20 psig and heat trace is not required, the annular space will be maintained at a vacuum and vacuum decay shall be used to monitor the annular space and alarm/interlock on loss of vacuum.
- j. For liquefied gases at a pressure > 20 psig and heat trace is required, the annular space will be maintained at 80 psig and pressure decay shall be used to monitor the annular space and alarm and interlock on loss of pressure.
- k. For liquefied gases at a pressure < 20 psig and heat trace may be required, the annular space will be maintained at 80 psig and pressure decay shall be used to monitor the annular space and alarm and interlock on loss of pressure.
- l. Where the annular space is to be maintained at 80 psig, the annular space will be pressurized with an inert gas (e.g, argon or helium) which is compatible with the process gas and acceptable with the manufacturing tool owner. Prior to pressurizing with the inert gas, the annular space shall be cycle purged to remove atmospheric air, oxygen, and moisture to prevent a hazardous reaction in the event of a leak in the inner containment tubing.

- m. For secondary containment pressure monitoring from gas cabinet to distribution (DVB or VMB), loss of pressure shall trigger an alarm for notification and investigation. This is meant to reduce production impact in the event of false alarm due to pressure switch fault or loss of pressure to atmosphere from fill port.
- n. For secondary containment pressure monitoring from VMB to tool, loss of pressure shall trigger interlock/shutdown of the affected VMB stick. Shutdown will only impact one tool in the event of a pressure switch fault or loss of pressure to atmosphere.
- o. Alternately for gas lines that require heat trace to prevent condensation, a single contained heat traced line can be deemed acceptable if the following requirements can be fulfilled and the exceptional case has been jointly reviewed and approved by site team, Global EHS, PSM, and GFTT:
  - i. Installation must utilize a single roll of 316L EP tubing with pre-installed heat trace.
  - ii. Welds and heat trace splices may only be installed within the source (gas cabinet), an exhausted gas box (VMB), and point of use (tool gas box). No welds or heat trace splices must be allowed between the source and tool.
  - iii. This exception is not applicable to non-heat trace gas lines.
- p. Line installation requires line purging and passivation.
- q. Approved single roll tubing: Swagelok Pre-Insulated Electric-Traced Tubing Bundle



## 8.15 Controller

- a. Gas Cabinet and BSGS control systems shall be PLC or microprocessor based with an LCD-HMI display. The controller shall continuously monitor system inputs and automatically perform purging operations by sequencing valve actuation. Adequate purging to be ensured by checking pressure and vacuum at each step within the purge cycles. The controller shall also have capability to shutdown the system and stop process gas flow if an unsafe condition arises.
- b. The LCD display shall provide a graphical display of the process gas panels (both left and right) with a visual indication of pneumatic valve positions, shutdown and fault alarms with time stamps, and controller status.
- c. Controller shall include an Emergency Stop (EMO) Button. [SEMI S2: 12]
  - i. Activation of the EMO button should de-energize all hazardous voltage and power greater than 240 volt-amps and power to the solenoids causing all of the pneumatic valves to close without removing power to the controller.
  - ii. Safety related devices (e.g., smoke detectors, gas detectors, pressure transducers) and controllers performing data/alarm logging and error recovery functions may remain energized.
  - iii. Resetting the EMO button should not re-energize circuits or equipment.
  - iv. The EMO circuit should require manual resetting so that power cannot be restored automatically.
  - v. The EMO button should be guarded to minimize accidental activation.
- d. Gas Cabinet and BSGS controllers shall have dual power supplies for connection to normal and emergency power for uninterrupted operation.
- e. Controller shall have a password security system to prevent unauthorized personnel from operating the equipment.
- f. Controllers shall have ethernet connectivity for communication with GMS.
- g. Controllers for HPM gases shall have connection for Z-purge with a pressure switch to monitor the pressure.
- h. Controllers shall provide the ability to force open pneumatic valves upon controller fault/failure.
- i. Controllers shall provide alarm upon loss of communication to SCADA system.
- j. Controllers shall come equipped with 20% spare I/O and 20% spare pneumatics.

- k. Controllers shall be provided with a means to automatically reboot/recover in the event of a communications failure without process interruption to gas flow.
- l. Any top or side penetrations into the controller enclosure must be water-proof fittings (NEMA 4 rated) to mitigate water leaks into the controllers as a result of gas room sprinkler activation.
- m. Controller shall be able to interface with a barcode reader system. The barcode system shall have the following capabilities:
  - i. Prevent the wrong cylinder hook-up by barcode scanning prior to cylinder change. There shall be an interlock control in place to prevent the connection of the wrong cylinder.
  - ii. Allow only trained and qualified persons to perform the cylinder change.
  - iii. Check for expiration and generate alarm when nearing the expiration date (user settable warning period).
  - iv. Perform lot tracking, consumption calculation, and inventory management.
  - v. Barcode readers to be wireless
  - vi. Barcode reader shall be explosion proof for flammable gases
  - vii. Barcode reader shall have high level password protection in place to bypass barcode scanning due to faulty barcode reader. Barcode bypass alarm shall be reflected in SCADA to alert unauthorized bypass and shall not reset until the barcode bypass has been removed.
- n. The controller shall be able to perform the following operational sequences/programs:
  - i. Pre-purge – controller program/sequence that is used to purge and evacuate the process gas from the process pigtail prior to cylinder change or pigtail maintenance.
  - ii. Cylinder change – controller program/sequence that assists personnel in changeout of the empty package. The package change shall be assisted by pre/post automatic purge sequences, and vacuum/pressure tests, with pre-set cycle time and acceptable parameters variance. The sequences shall be controlled by settable vacuum/pressure values of the pressure transducers.
  - iii. Post Purge – controller program/sequence that is used to purge and evacuate air and any atmospheric contaminants from the pigtail after the cylinder changeout or pigtail maintenance.
  - iv. Process Gas Flow – controller program/sequence that is used to initiate process gas flow.
  - v. Stop Process Gas Flow – controller program/sequence that is used to stop the on-line process gas flow.

- vi. Restart Process Gas Flow – controller program/sequence that is used to restart after stopping the flow of process gas.
- vii. Manual Mode – controller program for manual mode of operation that provides a means of actuating pneumatic valves outside of a programmed sequence. Manual mode provides a means of initiating the flow of purge gas through the purge and process gas lines during installation and pre-startup procedures, or during maintenance.
- viii. Auto-crossover – controller program/sequence that initiates automatic package switch-over from the “empty” package to the “full” package to ensure continuous supply of gas. The switch-over shall be automated by weight or pressure control, depending upon gas properties. For liquefied gases, the primary control for crossover shall be performed on the weight followed by package supply and delivery pressure. For non-liquefied gases, the primary control for crossover shall be performed by package supply and delivery pressure.
- ix. Digital input shall be provided to interlock the purge sequence in the HPM supply system connected to a local gas treatment/abatement system. This input shall disable and stop the purge sequence when the abatement system is inoperable (either in bypass or shutdown) to prevent the release of untreated gas into the exhaust ducts.

## 8.16 Gas Monitoring System

The Gas Monitoring System (GMS) shall be responsible for the monitoring (data acquisition) and auto shutdown notification of the gas distribution system, all its sub-assemblies, equipment, components, and communicate operational status information to the plant monitoring system (FMCS). The specialty gas delivery system shall have two levels of operation, local and remote.

- a. Local Operation: At the local level the operator can access/operate the gas cabinet controller located in the gas room to extract system operational information. The operator can stop/start gas flow or initiate purges. Audible and visual alarms are activated at the local controller if the system goes into an alarm condition.
- b. Remote Monitoring: At the remote level the operator can access informational data from the network to determine the status of any particular system
- c. Gas monitoring system shall be provided with emergency power.

## 8.17 Alarms

| <b>ALARM</b>                                      | <b>DESCRIPTION</b>                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Source Low Pressure</b><br>(GC & BSGS)         | On-line low source cylinder pressure fault alarm. This fault alarm provides notification that the container pressure is low. Low pressure means that the product will soon be depleted, but with liquefied compressed gas, can also indicate that the liquid has been sub-cooled due to a high withdrawal rate or low ambient temperature. |
| <b>Source Low-Low Pressure</b><br>(GC & BSGS)     | On-line low source cylinder pressure shutdown alarm. This alarm indicates an empty source container, a sub-cooled source container, excess flow, or a source cylinder PT failure.                                                                                                                                                          |
| <b>Source High Pressure</b><br>(GC & BSGS)        | On-line high source cylinder pressure fault alarm. If cylinder heating is provided, this alarm shuts down the cylinder heater. This alarm indicates a possible run-away source heater controller.                                                                                                                                          |
| <b>Purge Cycle Low Vacuum</b><br>(GC & BSGS)      | A shutdown alarm that prevents the purge sequence from finishing until sufficient vacuum is restored.                                                                                                                                                                                                                                      |
| <b>Purge High Pressure</b><br>(GC & BSGS)         | A shutdown alarm that prevents overpressure of process gas panel components in the event of a purge regulator failure. Prevents the purge sequences from finishing until the problem is resolved.                                                                                                                                          |
| <b>High Pressure Leak Test Low Purge Pressure</b> | A shutdown alarm during the high pressure leak test sequence (if HPLT is provided) when the HPLT purge gas pressure is low and insufficient to perform the                                                                                                                                                                                 |

|                                                                  |                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (GC & BSGS)                                                      | leak check. This alarm prevents the cylinder change sequence from finishing until the problem is resolved.                                                                                                                                                                                                                                 |
| <b>High Pressure Leak Test High-High Pressure</b><br>(GC & BSGS) | A shutdown alarm during the high pressure leak test sequence (if HPLT is provided) when the HPLT purge gas pressure is high. This prevents overpressure of the pigtail and cylinder connection in the event of the HPLT purge regulator failure. Prevents the cylinder change sequence from finishing until the problem is resolved.       |
| <b>Delivery Low Pressure</b><br>(GC & BSGS)                      | On-line low delivery pressure fault alarm. This fault alarm provides notification that the delivery pressure from the gas cabinet or BSGS is low. Low pressure typically means the regulator is set too low, or the flow rate has exceeded the system design, creating excessive pressure drop and regulator droop.                        |
| <b>Delivery Low-Low Pressure</b><br>(GC & BSGS)                  | On-line low-low delivery pressure shutdown alarm. This alarm may indicate excess flow, a closed regulator or valve, or delivery pressure transducer failure. This alarm will initiate shutdown of the on-line source/module and crossover to the standby source/module (provided a second source/module is available and in standby mode). |
| <b>Delivery High Pressure</b><br>(GC & BSGS)                     | High delivery pressure fault alarm. This alarm provides notification that the regulator supplying the houseline is creeping.                                                                                                                                                                                                               |
| <b>Delivery High-High Pressure</b><br>(GC & BSGS)                | High-High delivery pressure shutdown alarm. This alarm provides notification of a failed regulator. This alarm will initiate shutdown of the on-line source/module and crossover to the standby source/module (provided a second source/module is available and in standby mode).                                                          |
| <b>Vent Low Pressure</b><br>(GC & BSGS)                          | Low vent pressure fault alarm. This fault alarm may indicate low vacuum venturi N2 supply pressure, a closed vent valve, or a blocked or undersized vent line.                                                                                                                                                                             |
| <b>Vent High Pressure</b><br>(GC & BSGS)                         | High vent pressure fault alarm. This fault alarm may indicate a leaking purge gas isolation valve, manual process gas isolation valve left open, or a plugged vent line leading to the scrubber.                                                                                                                                           |
| <b>Coax Pressure Switch</b><br>(GC & BSGS)                       | The coax pressure switch monitors the annular space between the inner process pipe and the secondary containment. This alarm indicates that a leak has developed to atmosphere from secondary                                                                                                                                              |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | containment, or from the inner process pipe into the annular space.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Source Low Weight</b><br>(GC & BSGS)          | On-line low source cylinder weight fault alarm. This alarm provides notification that the cylinder weight is low, and product will soon be depleted. This is applicable for liquefied gases.                                                                                                                                                                                                                                        |
| <b>Source Low-Low Weight</b><br>(GC & BSGS)      | On-line Low-Low source cylinder weight shutdown alarm. This alarm indicates that the source cylinder is empty. This alarm initiates shutdown of the on-line cylinder and crossover to the opposite source cylinder if available. This is applicable for liquefied gases.                                                                                                                                                            |
| <b>Source High Weight</b><br>(GC & BSGS)         | On-line high source cylinder weight fault alarm. This alarm indicates that the scale may not be calibrated correctly, the net liquid weight is incorrect, or transfill is occurring between containers that are on line at the same time.                                                                                                                                                                                           |
| <b>Source High-High Weight</b><br>(BSGS)         | On-line high-high weight shutdown alarm. This shutdown alarm indicates that a transfill condition whereby two containers are on line at the same time, one cooler than the other, being filled by the warmer container. This alarm prevents overfill of a container resulting in an unsafe condition. This alarm will initiate shutdown of the online source and crossover to the opposite source if available and in standby mode. |
| <b>JT Heater Low Temperature</b><br>(BSGS)       | JT heater low temperature fault alarm. This alarm provides notification when there is power reduction or loss of power to the JT heater. Potential causes of this alarm include failed heater element, tripped circuit breaker, or momentary drastic increase in flow rate.                                                                                                                                                         |
| <b>JT Heater High Temperature</b><br>(BSGS)      | JT heater high temperature fault alarm. This alarm provides notification of an increase in temperature above the normal operating range of the heater. Potential causes of this alarm include temporary overshoot as the heater warms up during startup, or a momentary drastic decrease in flow rate.                                                                                                                              |
| <b>JT Heater High-High Temperature</b><br>(BSGS) | JT heater high-high temperature shutdown alarm. This alarm provides notification of a run-away JT heater and initiates shutdown of the heater by removing power to the heater. This alarm does not stop process gas flow or initiate a crossover.                                                                                                                                                                                   |

|                                              |                                                                                                                                                                                         |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Heat Trace Low Temperature</b><br>(BSGS)  | Fault alarm that detects a loss of power to the heat trace, tripped GFCB, failed contactor, or incorrect temperature setpoint.                                                          |
| <b>Heat Trace High Temperature</b><br>(BSGS) | Fault alarm that detects an increase in power to the heat trace.                                                                                                                        |
| <b>Low Exhaust</b><br>(GC & BSGS)            | Fault alarm that detects low exhaust flow.                                                                                                                                              |
| <b>Low Pneumatic Pressure</b><br>(GC & BSGS) | Fault alarm that detects low pneumatic pressure to the solenoid valves. If pneumatic pressure drops below 60 psig, process and purge valves become unreliable and may cause a shutdown. |
| <b>Low Z-Purge Pressure</b><br>(GC & BSGS)   | Fault alarm that detects low Z-purge pressure to the controller.                                                                                                                        |

#### 8.18 Safety Shutdowns and Interlocks

|                                                |                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Emergency Stop / EMO</b><br>(GC & BSGS)     | The emergency stop button is located on the controller. Activation of the EMO button will remove power from cylinder heating, heat trace, and all automatic valves will close and stop process gas flow.                                                                                                                                                                    |
| <b>Life Safety Shutdown</b><br>(GC & BSGS)     | This shutdown is the result of any Life Safety System (LSS) device, such as a gas detector sending a gas detect signal to Micron's TGMS which in turn sends a shutdown signal to this digital input to shut down the equipment. Upon receipt of this signal, power is removed from cylinder heating, heat trace, and all automatic valves close and process gas flow stops. |
| <b>Customer Remote Shutdown</b><br>(GC & BSGS) | Remote shutdown signal initiated from Micron FMCS. All automatic valves close and process gas flow stops.                                                                                                                                                                                                                                                                   |

#### 8.19 Gas Specific Requirements

##### a. Acetylene (C<sub>2</sub>H<sub>2</sub>)

- i. Acetylene decomposes rapidly in the gas phase with a considerable release of heat and a rapid increase in pressure. This hazard can be mitigated by delivering the gas at reduced pressure. Delivery pressure shall be < 15 psig.

- ii. Acetylene should not be delivered outside of the cylinder at a pressure in excess of 15 psig. To ensure this requirement, pressure reduction must occur directly at the outlet of the cylinder.
  - iii. Flashback arrestors shall be installed, preferably as close to the cylinder as possible.
  - iv. Common acetylene cylinders contain a porous filler material and a solvent (acetone or dimethylformamide, DMF). By using the porous material and solvent, it is possible for a cylinder to contain much more acetylene than could safely be compressed into the same cylinder without solvent. To minimize the withdrawal of liquid solvent, acetylene should be withdrawn from the cylinder at a rate not to exceed 1/10 (one-tenth) of the capacity of the cylinder per hour during intermittent use. For full withdrawal of the contents of the cylinder on a continuous basis, the flow rate should be no more than 1/15 (one-fifteenth) of the capacity of the cylinder per hour. Too high a flow rate causes solvent carryover.
  - v. Install pre-filter and final filter to protect against contamination from source cylinder.
  - vi. Although the pressure in the cylinder will decrease as acetylene is withdrawn, accurate measurement of the contents can only be obtained by weight. Weight scales shall be used as the primary mode of switchover, followed by cylinder pressure.
- b. Cylinder Separation for Pyrophoric Gases
- Individual cylinders containing pyrophoric gases shall be secured and separated from other pyrophoric gas cylinders to prevent flame impingement from a release to an adjacent cylinder or valve. Separation shall be by ¼ in (6 mm) thick steel plate extended a minimum of 18 in (460 mm) below the centerline of the cylinder valve and a minimum of 6 in (150 mm) above the centerline of the cylinder valve.
- c. Bulk Silane (SiH<sub>4</sub>)
- i. Provide optical flame detectors (UVIR) on silane delivery systems. Coverage shall be provided at container connections, process gas and purge gas panels, and other potential leak points where unwelded fittings or connections are used. [NFPA 318: 11.2.5.1; CGA G-13: 11.2.1]
  - ii. Automatic fixed deluge water spray protection shall be provided for bulk silane delivery systems. The system shall be designed to provide a density of 0.30 gpm/ft<sup>2</sup> over the external surface area of the package for a 2-hr duration. The water spray system shall be automatically activated by optical flame detectors (UVIR). [NFPA 318: 11.2.5.2.2; CGA G-13: 12.2.1]



- iii. BSGS for SiH<sub>4</sub> shall be provided with an automatic cylinder connection protection device to prevent unauthorized or unintended access to the cylinder connection.
- d. BSGS gases that come under Process Safety Management / Risk Management Plan Guidance

The Environmental Protection Agency's Risk Management Plan (RMP) Rule provides a [List of Regulated Substances](#) under Section 112(r) of the Clean air Act. This list shall be reviewed for BSGS gases, and scales are to be provided for those gases to ensure the maximum allowable quantity on site is not exceeded.

- e. HF for 21VE tools
  - i. No Hastelloy containing components (valves, regulators, filters, etc) are allowed in gas cabinets and VMBs supplying HF to 21VE tools.
  - ii. Do not install regulators in gas cabinets that supply HF to 21VE tools. Regulators will be installed in the VMB stick supplying the 21VE tool chamber. Provide a dedicated VMB stick to each 21VE tool chamber.
  - iii. Install a conical pre-filter in the HF gas cabinet. Do not install a final filter in the gas cabinet without review from TEL.
  - iv. Heat HF cylinders using a 1.5 kW electric blanket heater positioned at the bottom half of the cylinder, and wrap the entire cylinder with jacket insulation to retain heat.
  - v. Increasing temperature gradient from cylinder to the 21VE tool
    - HF cylinder temperature setpoint: 30°C
    - Gas cabinet panel, tubing from GC to VMB, and VMB temperature setpoint: 35°C
    - VMB Hookup tubing to tool's gas box temperature setpoint: 40°C
  - vi. Tool Hookup Temperature monitoring
    - For hookup length (VMB to gas box inlet) < 30 m, use one thermocouple for control positioned 15 cm away from gas box inlet.
    - For hookup lengths > 30 m, use two thermocouples for control with the second TC positioned 10 m away from the VMB.

## 8.20 Operation & Maintenance

- a. Specialty gases are generally distributed at delivery pressure < 100 Psig except for specific applications.
  - i. Compressed and compressed liquefied gases are typically distributed at  $\leq 75$  Psig.
  - ii. Liquefied low vapor pressure gases are distributed in a range between -10 Psig and 25 Psig, depending on the specific vapor pressure at room temperature.
- b. Specialty gases shall be supplied at room temperature, between 68 to 77°F (20 to 25°C), based on the site-specific operating parameters. However, in some applications, due to gas properties, the actual gas temperature may drop below 32°F with potential for condensation of the gas and ambient moisture condensing on the pressure regulator body. This effect shall be evaluated in the design phase and addressed with an adequate temperature control system (i.e., cylinder heating and/or heat trace of the process piping).
- c. Low vapor pressure gases may need to be operated with a dedicated temperature control system to increase capacity and prevent gas condensation within the distribution system.
- d. Gas supply systems shall be operated and maintained according to the manufacturer's operations and maintenance manuals as well as the site's and/or Total Gas Management (TGM) contractor's O&M procedures.
- e. Routine inspection and maintenance shall follow FCT Operations and Maintenance BKM Standard.

## 8.21 Capacity

The system capacity and concept design shall be identified based on, but not limited to, the following average flow categorization:

- a. Design flow  $\leq 10$  slpm
  - i. Design based on standard auto crossover gas cabinet/manifold.
  - ii. Typical indoor installation depending upon available package (cylinders)
  - iii. N+1 Redundancy for HPM gases, based upon gas properties and lessons learned. Individual gas panel for non-HPM gases.
  - iv. Temperature control as option for liquefied gases
- b. Design flow  $\geq 10$  slpm
  - i. Design based on pigtail module, high flow or BSGS system.
  - ii. Indoor or outdoor installation depending upon available package (Y-cyl, cylinder bundles, tube trailer, etc.)
  - iii. N+1 Redundancy for HPM gases, based upon gas properties and lessons learned. Individual gas panel for non-HPM gases
  - iv. Temperature control as option for liquefied gases. Redundancy/Reliability
- a. Gas supply systems shall be equipped with auto crossover manifold connected to a gas package to guarantee continuous supply.

Typical configuration is one side supplying gas and the other in stand-by, ready to supply.

- b. Gas supply system controllers shall be equipped with dual power supplies for connection to normal and UPS power sources to allow for continuous operation.
- c. For supply of critical HPM gases, supply and distribution systems may be configured as Configurations 6.4.d or 6.4.f with N+1 design.
- d. For BSGS supply, each fab building shall have it's own dedicated BSGS system including backup. The BSGS system may be configured as 6.4.e. A BSGS system (N) shall be equipped with an auto crossover manifold with at least two packages (A & B) where one package is connected to each side. The backup system (+1) may be a single manifold connected to a single package (C).

## 8.22 Expansion

- a. All gas supply systems shall be provided with two process outlets as minimum with following exceptions:
  - i. Point/point distribution systems (Configuration 6.4.a)
  - ii. Specialized gas utilized for metrology tools or other limited applications

## 8.23 Approved Equipment Vendors

- a. Versum
- b. Air Liquide
- c. Collabratech Solutions
- d. KCTech
- e. Taiyo Nippon Sanso Taiwan (TNST)
- f. Toyoko (TYK)
- g. Ceres
- h. Diversified Fluid Solutions (DFS)

## 8.24 Factory Certification

Spec gas equipment shall be certified to the minimum specifications, or better, prior to delivery to the site:

- a. All high purity gas panels shall be certified to the following purity specifications at a minimum:

|           |                                               |
|-----------|-----------------------------------------------|
| Oxygen    | ≤50 ppb above background                      |
| Moisture  | ≤50 ppb above background                      |
| Particles | ≤10 @ > 0.1 micron/scf, 3 consecutive counts  |
|           | ≤50 @ > 0.02 micron/scf, 3 consecutive counts |

- b. All gas panels will be tested to the following leak rates:

|             |                              |
|-------------|------------------------------|
| Inboard     | 1 x 10 <sup>-9</sup> sccs He |
| Outboard    | 5 x 10 <sup>-6</sup> sccs He |
| Across Seat | 5 x 10 <sup>-6</sup> sccs He |

- c. All systems will be shipped pressurized with Argon to the last valve at 20 psig. Tube stubs will be double bagged using cleanroom baggies. Certification and QC documentation will be provided with each system.

## 8.25 Factory Acceptance Tests

- a. Prior to shipping, manufacturer/supplier shall perform factory acceptance testing on spec gas equipment including but not limited to the following:

i. Panel QC Testing

- Pressure test
- Helium leak test
- Moisture & oxygen analysis
- Particle Counts

ii. Controller Function Test

- Digital Input Tests
- Pneumatic Valve Tests
- Analog Signal Tests
- Analog Alarm Tests

iii. Visual inspection and verification against PFD/P&ID

- b. Equipment manufacturer shall provide a Factory Acceptance Test Report at shipment. See Appendix 2 for Factory Acceptance Test Documentation Example Form.

## 8.26 Equipment Supplier Documentation

The equipment supplier shall provide manuals and documentation based on the originally intended use of the equipment. The manuals and related documentation shall describe the scope and normal use of the equipment, and provide information to enable safe facilitation, operation, maintenance, and service of the equipment. All documents are to be provided in both hardcopy and digital formats. Documentation and manuals should include:

- a. Specific instructions on operations and routine tasks (Sequence of Operations);
- b. Instructions for energy isolation (COHE);
- c. Description of the emergency off (EMO) and interlock functions;
- d. Maintenance and troubleshooting procedures needed to maintain the effectiveness of safety design features or devices;
- e. Maintenance procedures with potential environmental impacts – identify procedural steps during which releases might occur, and identify methods to minimize the volume of effluents or emissions generated during maintenance procedures;
- f. System overview (P&ID, PFD, etc.);
- g. Component/Material list with OEM part numbers;
- h. Controller features overview including one-line drawings, O&M manual, component/material list with OEM part numbers;
- i. Controller memory map with PLC/software program and unlocked software for HMI program for integration with GMS system;
- j. Factory Certification Test reports;
- k. Factory Acceptance Test reports

## 8.27 Installation

- a. Gas supply systems shall be installed by personnel specifically trained and certified for high purity systems, including the welding and assembly, and testing and qualification processes.
- b. For installations performed by external companies, those companies shall provide all documentation related to the training and certification of all personnel involved in welding, assembly, testing, and qualification processes of stainless steel high purity systems using the execution requirements contained in the "UHP Stainless Steel BKMs guidelines" document.

## 8.28 Pipeline QA / Testing

- a. Gas supply systems shall be finally qualified based on, but not limited to, the following tests:
- b. Leak test and/or pressure test to certify the mechanical integrity of welding and VCR connections. The entire system shall be subjected to a pressure test at a minimum pressure of 20 percent over the maximum pressure available to the system but not less than 80 psig for 24 hours with no discernable pressure drop.
- c. Moisture content to certify the complete dry out of the interior of the pipeline to avoid any abnormal reaction with the specialty gas.
  - i. Particle test to certify the cleanliness and good quality of the welds and ensure that any particle contamination does not exceed the maximum accepted level.

## 9.0 Inter-Discipline Coordination Matrix:

Site gas construction teams and GFTT to engage the appropriate site construction disciplines during the design phase to ensure the provisions are addressed.

### Provisions Provided by Each Discipline

| Discipline    | Provision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Architectural | <ul style="list-style-type: none"><li>• HPM Rooms and Gas Rooms shall be separated from other areas by not less than a 2-hr fire barrier where the area is 300 square feet or more and not less than a 1-hr fire barrier where the area is less than 300 square feet.</li><li>• Rooms containing pyrophoric gases shall be constructed of 8" concrete walls. A 4-hr fire resistive concrete wall shall be constructed between the gas room and adjacent H-5 occupancies.</li><li>• HPM Rooms, Gas Rooms, and liquid Storage Rooms, shall have at least one exterior wall and such wall shall be not less than 30 ft from property lines, including property lines adjacent to public ways.</li><li>• HPM rooms, gas rooms and liquid storage rooms shall have two exits. One exit shall be directly to the outside of the building.</li><li>• Install normally closed or automatic closing FM Approved fire doors with 1.5 hr fire rating in fire-rated walls.</li><li>• Provide Damage-Limiting Construction in accordance with FM Global Data Sheet 1-44, "Damage-Limiting Construction" where applicable.</li><li>• Rooms containing water reactive materials must be resistant to water penetration and liquid plumbing cannot be present excluding fire sprinklers.</li><li>• Provide adequate fire stopping at all required fire wall penetrations</li><li>• Lighting shall provide 60 ft-candle coverage for normal working areas within the gas room</li><li>• Drains shall be designed to provide adequate drainage for all fire sprinkler systems, safety shower/eye wash stations, etc.</li><li>• Emergency shower and eye wash stations, safety showers and eyewashes shall be placed in the each HPM-Gas Room and each Gas Utility Room.</li></ul> |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mechanical         | <ul style="list-style-type: none"><li>• Provide room mechanical exhaust ventilation at a minimum rate of 0.31 m3/min per m2 of floor area (1 ft3/min per ft2).</li><li>• Design gas cabinet exhaust ventilation systems to operate at not less than one-half the normal fan speed on the emergency power system where it is demonstrated that the level of exhaust will maintain a safe atmosphere.</li><li>• Exhaust ventilation from gas cabinets and exhausted enclosures shall be directed to a treatment system</li><li>• Maintain room temperature at 70 + 2 °F<br/>For Toxic, Highly Toxic, and Pyrophoric compressed gases, follow IFC 6004.1.2 to provide average face velocity at the gas cabinet's access port at no less than 200 fpm with a minimum of 150 fpm at any point.</li></ul>                                                                                                                                                                          |
| Electrical         | <ul style="list-style-type: none"><li>• Provide grounding for all Electrical equipment/components</li><li>• Provide GFCI for cylinder heating and heat trace power.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Water / Wastewater | <ul style="list-style-type: none"><li>• Provide drain for water from safety showers, eye wash stations, and fire sprinklers</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| I&C                | <ul style="list-style-type: none"><li>• HPM Rooms, Gas Rooms and Liquid Storage Rooms shall be equipped with an approved emergency alarm system, and emergency alarm initiating devices shall be installed outside of each interior exit door of such rooms.</li><li>• Access into HPM Rooms, Gas Rooms, and Liquid Storage Rooms shall be controlled with a card reader.</li><li>• Gas cabinet exhaust shall be monitored and alarm to the SCADA in case of an exhaust system failure</li><li>• All safety shower and eyewash stations will be installed with a drain, activation monitoring, and alarm back to the control room in areas that are not typically occupied spaces and have significant safety and health hazards</li><li>• Emergency alarms (horns and strobes per code and Micron policy) shall be designed into the gas rooms</li><li>• Provide FMCS/SCADA system monitoring and controls.</li><li>• Provide LSS system monitoring and controls.</li></ul> |



## 10.0 Startup Commissioning / Site Acceptance Test

- a. For all new equipment installations, follow Global EHS – Equipment Installation Standard, Document No. 2W4373RQWREN-1568922467-12.
- b. At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Commissioning / Site Acceptance Test:
  - i. Visual Inspection to assess overall condition and identify any damage during installation.
  - ii. Mechanical installation verification
  - iii. Electrical installation verification
  - iv. Labeling
  - v. Functional testing of digital inputs/outputs, alarm verification, and validation at Micron SCADA.
  - vi. Functional testing of analog inputs, alarm verification, and validation at Micron SCADA.
- c. See Appendix 3 for Startup and Commissioning Documentation Example Form.
- d. ClF3, HF, F2 mixes and fluoride supply systems shall be passivated to allow the electro-polished stainless steel wetted surface to form a metal oxide and fluoride layer.

## 11.0 Document Control

### 11.1 Approval: GLOBAL\_FAC\_GFTT\_APPR

### 11.2 Notification:

GLOBAL\_FAC\_MANAGERS  
GLOBAL\_FAC\_COMMUNICATIONS  
GLOBAL\_FAC\_CONSTRUCTION  
GLOBAL\_FAC\_NOTIFY  
FCG\_F2\_FAC\_NOTIFY  
FCG\_F4\_FAC\_NOTIFY  
FCG\_F6\_FAC\_NOTIFY  
FCG\_F10\_FAC\_NOTIFY  
FCG\_F11\_FAC\_NOTIFY  
FCG\_F15\_FAC\_NOTIFY  
FCG\_F16\_FAC\_NOTIFY  
FCG\_MMP\_FAC\_NOTIFY  
FCG\_MMY\_FAC\_NOTIFY  
FCG\_MSB\_FAC\_NOTIFY  
FCG\_MXA\_FAC\_NOTIFY  
FCG\_MTB\_FAC\_NOTIFY

### 11.3 Final Viewer Access

All Micron Team Members at all Sites

### 11.4 Retention

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

### 11.5 Review

Biennially (two years)

### 11.6 Revision History

| Revision # | Section Changed | Details of Changes | Date     |
|------------|-----------------|--------------------|----------|
| 0          | NA              | New document       | 03/01/21 |

12.0 Appendix

Appendix 1: Template Form for Site Specific Deviation List

Site: \_\_\_\_\_

| No | Specification Section and Item | Deviation Proposal | Scope Impact (Benefit, Cost, Schedule) | Remark | Approval:<br>1. Site<br>2. GFTT<br>3. FCT |
|----|--------------------------------|--------------------|----------------------------------------|--------|-------------------------------------------|
|    |                                |                    |                                        |        |                                           |
|    |                                |                    |                                        |        |                                           |
|    |                                |                    |                                        |        |                                           |
|    |                                |                    |                                        |        |                                           |

**APPENDIX 2:****FACTORY ACCEPTANCE TEST DOCUMENTATION (Example form)**  
**TEMPLATE FOR FACTORY ACCEPTANCE TEST**

(Manufacturer may provide their own report and format, but all items must be addressed)

**GAS CABINET / BSGS****GC & BSGS Process Panel Testing**

|                     |  |          |        |
|---------------------|--|----------|--------|
| Manufacturer        |  | Customer | Micron |
| GC, BSGS Model      |  | Gas      |        |
| GC, BSGS ID         |  |          |        |
| Micron Equip ID     |  |          |        |
| P&ID No.            |  |          |        |
| Panel ID/Serial No. |  |          |        |

**PUNCH LIST (Items to be completed at site):**

| No. | Punch List Item | Date | Owner | Closure of Punch List |      |
|-----|-----------------|------|-------|-----------------------|------|
|     |                 |      |       | Date                  | Name |
|     |                 |      |       |                       |      |
|     |                 |      |       |                       |      |

**PANEL QC TESTING (PRESSURE, HELIUM LEAK, MOISTURE, OXYGEN, PARTICLES)****A. PRESSURE TEST**

|                                   |                               |       |                               |
|-----------------------------------|-------------------------------|-------|-------------------------------|
|                                   | Date                          | Time  | Pressure (psig)               |
| Start Test                        |                               |       |                               |
| End Test                          |                               |       |                               |
| Pressure Difference (Start - End) |                               |       |                               |
| Acceptance Criteria               | Test Pressure $\geq$ 100 psig |       | <input type="checkbox"/> PASS |
|                                   | Test Duration > 4 hrs         |       | <input type="checkbox"/> FAIL |
|                                   | $\Delta P = 0$ psig           |       |                               |
| Inspector Name                    |                               | Date: |                               |

**Include copy of pressure trend****B. INBOARD HELIUM LEAK TEST****a. Helium Leak Detector Used**

|              |           |            |
|--------------|-----------|------------|
| Manufacturer | Model No. | Serial No. |
|              |           |            |
|              |           |            |

Last Modified: 03/01/21

Version: 1

ID: A3YRXSD74VDV-57553043-413

## b. Inboard Helium Leak Test

| Item                                                          | Result                             |       |      |
|---------------------------------------------------------------|------------------------------------|-------|------|
|                                                               | NA                                 | PASS  | FAIL |
| Pump down panel to desired vacuum, then perform He leak check |                                    |       |      |
| He Leak Test Criteria                                         | $\leq 1 \times 10^{-5}$ atm cc/sec |       |      |
| Inspector Name:                                               |                                    | Date: |      |

## C. MOISTURE TEST

## a. Moisture Analyzer Used

| Manufacturer | Model No. | Serial No. |
|--------------|-----------|------------|
|              |           |            |
|              |           |            |

## b. Moisture Test

|                      |               |         |  |
|----------------------|---------------|---------|--|
| Date:                |               | Time:   |  |
| Acceptance Criteria: | $\leq 10$ ppb | Result: |  |
| Inspector:           |               | Date:   |  |

**Include copy of moisture trend**

## D. OXYGEN TEST

## a. Oxygen Analyzer Used

| Manufacturer | Model No. | Serial No. |
|--------------|-----------|------------|
|              |           |            |
|              |           |            |

## b. Oxygen Test

|                      |               |         |  |
|----------------------|---------------|---------|--|
| Date:                |               | Time:   |  |
| Acceptance Criteria: | $\leq 10$ ppb | Result: |  |
| Inspector:           |               | Date:   |  |

**Include copy of oxygen trend**

## E. PARTICLE TEST

## a. Particle Counter Used

| Manufacturer | Model No. | Serial No. |
|--------------|-----------|------------|
|              |           |            |
|              |           |            |

## b. Particle Test

|                      |                                                   |         |  |
|----------------------|---------------------------------------------------|---------|--|
| Date:                |                                                   | Time:   |  |
| Acceptance Criteria: | $\leq 1$ particle @ $0.1 \mu\text{m}/\text{ft}^3$ | Result: |  |
| Inspector:           |                                                   | Date:   |  |

**Include copy of particle trend**

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**CONTROLLER FUNCTION TEST**

## A. Controller Configuration

| Controller Type (PLC, Microprocessor) | Controller Manufacturer / Model No. | Serial Number |
|---------------------------------------|-------------------------------------|---------------|
|                                       |                                     |               |

| Item                | Version | Serial Number |
|---------------------|---------|---------------|
| Controller Firmware |         |               |
| Controller Software |         |               |
| HMI                 |         |               |

## B. Digital Input Test

| Input                      | Test |     |         |           | Result                                                                                        |
|----------------------------|------|-----|---------|-----------|-----------------------------------------------------------------------------------------------|
| Gas Leak Shutdown          | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Temperature Sensor         | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Exhaust Switch             | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Low Pneumatic Pressure     | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Z-Purge Switch             | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Remote Shutdown            | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| EMO Shutdown               | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Excess Flow Switch – Left  | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Excess Flow Switch – Right | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass                                  |

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|                              |     |     |         |           |                                                                                                                                |
|------------------------------|-----|-----|---------|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| Coax Pressure Switch – Left  | HMI | LED | Audible | Reset/ACK | <input type="checkbox"/> Fail<br><input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| Coax Pressure Switch – Right | HMI | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail                                  |

| Others (To Be Filled in by Supplier) |      |     |         |           |                                                                                               |
|--------------------------------------|------|-----|---------|-----------|-----------------------------------------------------------------------------------------------|
| Input                                | Test |     |         |           | Result                                                                                        |
|                                      | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
|                                      | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
|                                      | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
|                                      | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
|                                      | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
|                                      | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA<br><input type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

## C. Valve Test

(List all pneumatic valves to be tested and check result)

| Right Panel Valves |    |      |      | Left Panel Valves |    |      |      |
|--------------------|----|------|------|-------------------|----|------|------|
| Valve              | NA | PASS | FAIL | Valve             | NA | PASS | FAIL |
|                    |    |      |      |                   |    |      |      |
|                    |    |      |      |                   |    |      |      |
|                    |    |      |      |                   |    |      |      |
|                    |    |      |      |                   |    |      |      |

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## D. Analog Signal Test

(List all pressure transducers and check result)

| PT ID | Test                           | Verification                                                             | Result                      |                               |                               |
|-------|--------------------------------|--------------------------------------------------------------------------|-----------------------------|-------------------------------|-------------------------------|
|       | Set PT to atmospheric pressure | Check for zero psig reading on PT, switch, and gauge                     | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       | Set PT to > 100 psig           | Check for reference pressure.<br><br>Reference Pressure<br>( _____ psig) | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|       |                                |                                                                          | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |

(List all weight scales and check result)

| Scale ID | Test                             | Verification                                                         | Result                      |                               |                               |
|----------|----------------------------------|----------------------------------------------------------------------|-----------------------------|-------------------------------|-------------------------------|
|          | No weight on scale (adjust ZERO) | Check for zero weight reading                                        | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|          |                                  |                                                                      | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|          |                                  |                                                                      | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|          | Check against known weight       | Check for reference weight<br>Reference Weight:<br>( _____ lbs / kg) | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|          |                                  |                                                                      | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|          |                                  |                                                                      | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |



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## E. Analog Alarm Test

| Alarm                              | Test |     |         |           | Result                      |                               |                               |
|------------------------------------|------|-----|---------|-----------|-----------------------------|-------------------------------|-------------------------------|
| Left Cylinder Pressure Low Alarm   | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Left Cylinder Pressure High Alarm  | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Right Cylinder Pressure Low Alarm  | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Right Cylinder Pressure High Alarm | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Left Delivery Pressure Low Alarm   | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Left Delivery Pressure High Alarm  | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Right Delivery Pressure Low Alarm  | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Right Delivery Pressure High Alarm | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Left Scale Low Alarm               | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Left Scale Low-Low Alarm           | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Right Scale Low Alarm              | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
| Right Scale Low-Low Alarm          | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|                                    | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|                                    | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |
|                                    | HMI  | LED | Audible | Reset/ACK | <input type="checkbox"/> NA | <input type="checkbox"/> Pass | <input type="checkbox"/> Fail |

**APPENDIX 3:****STARTUP AND COMMISSIONING (Example Form) Visual Inspection & General Condition**

| Item                                                                                         | Punch List | Responsible | Initial / Date |
|----------------------------------------------------------------------------------------------|------------|-------------|----------------|
| All component / equipment conditions OK                                                      |            |             |                |
| All mechanical (VCR / Swagelok) connections are tight                                        |            |             |                |
| All equipment (scales, heat trace, heat trace controller, cylinder heaters, etc.) installed. |            |             |                |
| Roof / canopy for outdoor BSGS installed and complete (if applicable)                        |            |             |                |

**A. Mechanical Installation**

| Item                                                                                                                                                                                                               | Punch List | Responsible | Initial / Date |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|----------------|
| Verify all equipment properly mounted and secured to floor/concrete pad (anchor bolts), supports, or walls. Verify seismic bracing installed if required.                                                          |            |             |                |
| All piping (process, purge, vent, pneumatic, venturi, etc.) has been flow checked and verified as accurate against PFD / P&ID.<br>Dwg# _____, Rev. _____                                                           |            |             |                |
| Verify all piping and components properly mounted and supported on GC/BSGS backplate. Verify all pipe clamps are tightened.                                                                                        |            |             |                |
| Verify all bulkhead fittings on piping are tight.                                                                                                                                                                  |            |             |                |
| Verify all doors close and latch properly, check alignment, and seals/gaskets are not damaged.                                                                                                                     |            |             |                |
| Verify door locks function properly.                                                                                                                                                                               |            |             |                |
| Process lines from GC or BSGS installed and testing complete (pressure test, helium leak test, O <sub>2</sub> , moisture, and particles)                                                                           |            |             |                |
| Vent lines from GC or BSGS installed and testing complete. Verify vent lines are installed to correct header and all fittings are tight. If vent header is purged, ensure purge source is connected and available. |            |             |                |
| Pneumatic tubing is installed and has been verified as accurate against the installation drawings.                                                                                                                 |            |             |                |

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| Item                                                                                                                                                                                                         | Punch List | Responsible | Initial/Date |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|--------------|
| Pneumatic supply is connected to each GC / BSGS and 85-95 psig is available.                                                                                                                                 |            |             |              |
| Verify vacuum venturi N2 supply line is installed, tested, and 85-95 psig of N2 is available.                                                                                                                |            |             |              |
| Verify Z-purge supply line installed, tested, and 85-95 psig of N2 is available                                                                                                                              |            |             |              |
| Verify exhaust duct / scrubber exhaust lines are connected, flowing, and balanced (if applicable).                                                                                                           |            |             |              |
| Verify interconnect piping from GC/BSGS to downstream module (VMB/MVB/DVB) tested and complete (pressure test, helium leak test, O2, moisture, and particles). Verify exhaust flow tell-tails are installed. |            |             |              |
| Verify heat trace and insulation is installed (if applicable) from GC/BSGS to downstream module.                                                                                                             |            |             |              |
| Hazardous gas monitoring/leak detection installed, wired, and tested (if applicable).                                                                                                                        |            |             |              |
| For BSGS with Y-Cylinder Scale, verify scale is properly mounted, including seismic bracing if required, properly wired, and terminal box is weather-tight.                                                  |            |             |              |
|                                                                                                                                                                                                              |            |             |              |
|                                                                                                                                                                                                              |            |             |              |

**B. Purge System**

| Item                                                                                                                                                                                                                                                                                                                    | Punch List | Responsible | Initial / Date |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|----------------|
| Verify purge lines installed and tested. Verify all connections are tight.                                                                                                                                                                                                                                              |            |             |                |
| Verify purge and helium leak check cylinders are available and connected to purge manifolds.                                                                                                                                                                                                                            |            |             |                |
| Purge gas purifiers installed (if applicable)                                                                                                                                                                                                                                                                           |            |             |                |
| Verify Helium leak check is installed (if applicable) and leak tested. For HPLT option, the leak check pressure should be regulated to 950 psig and have a safety relief valve set at 1200 psig. For a low pressure system, the leak check should be regulated to 200 psig and the safety relief valve set at 250 psig. |            |             |                |

**C. Electrical Installation**

| Item                                                                                                                                                                                              | Punch List | Responsible | Initial / Date |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|----------------|
| Verify all field conduit and wiring installed per design package and terminated at both ends.                                                                                                     |            |             |                |
| Electrical power connected.                                                                                                                                                                       |            |             |                |
| I/O signal wiring to GMS or SCADA installed.<br>Dwg# _____ Rev. _____                                                                                                                             |            |             |                |
| Ethernet network cable installed and connected                                                                                                                                                    |            |             |                |
| Conduit seals installed (if applicable), packed, and poured after electrical checkout complete and all wiring verified.<br>Tag/identify conduit seals once poured.                                |            |             |                |
| Life Safety System (LSS) I/O installed, wired, labeled, and point-to-point tests complete                                                                                                         |            |             |                |
| Verify that equipment is labeled with nameplate indicating source of incoming power feeds, voltage level, current, and frequency.                                                                 |            |             |                |
| For outdoor installations, verify all conduits weather-tight with o-ring gaskets, and low-point water drains provided.                                                                            |            |             |                |
| Ensure all circuit breakers and switches are in the Open/Off position. Verify all internal wiring is tight and properly terminated.                                                               |            |             |                |
| Verify incoming power feeds with electricians per installation drawing.                                                                                                                           |            |             |                |
| Verify Megger Testing (insulation resistance test) for incoming power feeds complete, and test results for wires/cables are in good condition.                                                    |            |             |                |
| Test and record incoming power feeds with electricians at power panel.                                                                                                                            |            |             |                |
| If located outdoors, verify that equipment have lightning protection.                                                                                                                             |            |             |                |
| Verify that equipment grounding conductors are connected and perform ground resistance test for each grounding conductor. Record test results of ground resistance with reference to ground grid. |            |             |                |

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**D. Labeling**

| Item                                  | Punch List | Responsible | Initial / Date |
|---------------------------------------|------------|-------------|----------------|
| HMIS labels installed for correct gas |            |             |                |
| Process piping correctly labeled      |            |             |                |
| Utility piping correctly labeled      |            |             |                |
| GC / BSGS correctly labeled           |            |             |                |
| Valve tags/labels installed           |            |             |                |
|                                       |            |             |                |

**E. Functional Test for Digital Inputs and Outputs**

| Item                                                                                                                                                                                                                                                                                                        | Punch List | Responsible | Initial / Date |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|----------------|
| Record drawing number and revision for Alarm Matrix.<br>Dwg# _____<br>Rev. _____                                                                                                                                                                                                                            |            |             |                |
| As each Fault Alarm or Shutdown Alarm is activated on the controller, verify the Relay Outputs are causing the correct response. Also, verify the outputs to the Micron SCADA.                                                                                                                              |            |             |                |
| From the controller, manually actuate each automatic valve individually. Perform point-to-point checks on all pneumatic tubing. Verify that each pneumatic line opens the correct valve. Verify operation of each valve by listening for valve venting and verify graphic change on the controller display. |            |             |                |
| Verify heaters (i.e., cylinder heating, Y-cyl heating, JT-heaters, heat tracing, etc.) are properly connected and functioning.                                                                                                                                                                              |            |             |                |
| For each "heater", take an Ammeter reading on the power wiring and ground wiring, and record the full-load current and the ground current.                                                                                                                                                                  |            |             |                |
| Test each Circuit Breaker and verify loss of voltage. Reset circuit breaker.                                                                                                                                                                                                                                |            |             |                |
| Using a 4-20 mA simulator, connect to the appropriate analog input and verify all fault and shutdown setpoint/alarms for each "heater" according to the Alarm Matrix.                                                                                                                                       |            |             |                |

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| Item                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Punch List | Responsible | Initial/Date |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|--------------|
| Activate the Emergency Stop push button on the controller and verify that the "Emergency Stop" Shutdown Alarm is displayed. Pull out the Emergency Stop push button to reset, acknowledge the "Reset", and verify that the alarm is cleared.                                                                                                                                                                                                                                                                     |            |             |              |
| Verify Low Z-Purge Pressure Fault Alarm. Re-establish Z-purge and verify that alarm can be cleared. Verify the controller is sealed by achieving and maintaining Z-purge pressure within the controller enclosure.                                                                                                                                                                                                                                                                                               |            |             |              |
| Verify Pneumatic Pressure alarms. Return pneumatic pressure to normal range, and verify that the alarm can be cleared.                                                                                                                                                                                                                                                                                                                                                                                           |            |             |              |
| Verify High Temperature Switch Shutdown Alarm. Verify that the High Temperature Shutdown alarm activates on the controller when the temperature switch is activated. Reset the temperature switch and verify that the alarm can be cleared.                                                                                                                                                                                                                                                                      |            |             |              |
| Verify UVIR flame detectors are properly configured for the appropriate gas type. Verify that the UVIR Fault alarm activates on the controller when the UVIR detector is in a fault condition. Verify that the UVIR Flame Verify shutdown alarm activates on the controller when the UVIR detector is in a shutdown condition. Verify proper response according to Alarm Matrix (i.e., "heaters" may be de-energized during a flame detect). Reset UVIR detector alarm and verify that the alarm can be cleared. |            |             |              |
| Verify Coax Pressure Switch alarm. Re-establish coax pressure and verify that the alarm can be cleared.                                                                                                                                                                                                                                                                                                                                                                                                          |            |             |              |

| Item                                                                                                                                                                                                                                                                                                 | Punch List | Responsible | Initial/Date |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|--------------|
| Verify the "Low Exhaust" fault alarm activates on the controller when the exhaust flow is reduced. Re-establish exhaust and verify that the alarm can be cleared.                                                                                                                                    |            |             |              |
| Verify that the "Life Safety System" Shutdown alarm activates on the controller when the LSS signal is de-energized. Verify proper response according to the Alarm Matrix (i.e., heaters may be de-energized during an LSS shutdown). Reset the LSS signal and verify that the alarm can be cleared. |            |             |              |
| Verify that the "Remote Shutdown" alarm activates on the controller when the remote shutdown signal from Micron's SCADA is de-energized. Reset the signal and verify that the alarm can be cleared.                                                                                                  |            |             |              |
| Verify that the "Vent/Scrubber Unavailable" Shutdown alarm activates on the controller when the Vent/Scrubber Unavailable signal is de-energized. Reset the signal and verify that the alarm can be cleared.                                                                                         |            |             |              |

**F. Functional Tests for Analog Inputs**

| Item                                                                                                                                                                                                                                                                    | Punch List | Responsible | Initial / Date |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|----------------|
| Verify that the setpoints in the controller match the Alarm Matrix                                                                                                                                                                                                      |            |             |                |
| Verify the purge parameters and leak check setpoints in the controller match the Alarm Matrix                                                                                                                                                                           |            |             |                |
| Verify the Alarm Responses, Conditions, and Delays in the controller match the Alarm Matrix                                                                                                                                                                             |            |             |                |
| Verify each pressure and/or weight alarm listed in the Alarm Matrix is tested, and the appropriate fault or shutdown alarm is activated on the controller. Verify the corresponding Relay Output causes the appropriate response. Verify the Outputs to Micron's SCADA. |            |             |                |

**G. Closeout**

| Item                                                                                                                                                         | Punch List | Responsible | Initial / Date |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|----------------|
| As-built / redlined drawings updated and put into Operating Manual                                                                                           |            |             |                |
| Copies of all documents have been provided to Micron Facilities Gas Engineer                                                                                 |            |             |                |
| Copy of the Startup and Commissioning document signed off by responsible persons and provided to Micron Facilities Gas Engineer                              |            |             |                |
| Verify that all Operating Manuals, keys, safety documentation, and commissioning spare parts are handed over to Micron Facilities Gas Engineer or designate. |            |             |                |